Social Preferences as Human Capital: Intergenerational Transmission in an Ethnically Diverse Context

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Abstract

Prosociality is essential for cooperation and the functioning of human societies, and is associated with important economic outcomes such as income, labor market prospects, or welfare. Social preferences are a key determinant of prosocial behavior. Yet, despite their central role in shaping behavior, relatively little is known about how these preferences are formed and what factors shape them. In this paper, I study how ethnic identity and parental influence are associated with social preferences. To do so, I elicit social preferences across 842 parent-child pairs from both ethnic majority and Roma minority in Bulgaria using survey- and task-based items. The results indicate that prosocial behavior and a preference for the in-group are common in this context and are positively associated between generations. I also find that in-group preferences are driven by a relatively small number of individuals across groups, as a majority of parents and children do not display an in-group or out-group preference. I also explore potential mechanisms associated with the intergenerational correlation of social preferences. In ethnic majority households, positive stereotypes, daily contact with the out-group, and a prosocial parenting style are associated with the intergenerational correlation of social preferences. Conversely, in ethnic Roma households, only a prosocial parenting style is associated with this intergenerational correlation.

1 Introduction

Prosocial behavior facilitates cooperation and collective action and is associated with economically important outcomes later in life, including labour-market success, income, and welfare (Algan et al., 2022; Deming, 2017; Kosse & Tincani, 2020). An important determinant of such behavior is social preferences: individuals' willingness to incur a cost to improve others' material payoffs (Fehr & Charness, 2025). Evidence that social preferences and prosocial behavior are malleable during childhood raises the possibility that prosociality can be developed over time (Algan & Huillery, 2025; Cappelen et al., 2020; Kosse et al., 2020).

Therefore, given their malleability and economic relevance, it is important to understand better what determines social preferences, especially during early life stages. In recent years, several studies have shown that cultural, familiar, and environmental factors can influence individual economic preferences

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(Ertac, 2020; Giuliano & Spilimbergo, 2025). However, there are still a lot of open questions regarding what and how economic preferences are determined (Ertac, 2020). For instance, an underexplored aspect is the interaction between specific determinants in shaping economic preferences. For example, two well-established facts in the social preferences literature are that group identities influence social preferences (Chen & Li, 2009; Kranton et al., 2020) — as we tend to be more prosocial with those that are like us (Balliet et al., 2014; Lane, 2016; Li, 2020) — and that children's social preferences tend to positively correlate with their parents' (Chowdhury et al., 2022; Dohmen et al., 2012). But we do not know if (and how) these two aspects interact with each other in the determination of social preferences. Do children's social preferences correlate with their ethnicity and their parents' social preferences? Does this intergenerational association correlate with the household's ethnicity? If so, what potential mechanisms may explain why? Answering these questions would provide a better grasp of what determines social preferences during childhood, enabling, for example, a better design of policies aimed at strengthening prosocial behaviors among children and adults.

Therefore, in this paper I estimate how ethnic group identity and parental preferences are associated with children's social preferences. To do so, I implemented a lab-in-the-field experiment and a survey in 842 Bulgarian households, with approximately 40% of them being ethnic Roma, to elicit the social preferences of a child and one of their parents. In each household, both the child and the parent played as the sender in two non-incentivized dictator games to elicit their altruism towards others. For both generations, I consider that displaying altruism towards others is a prosocial behavior and an indicator of their social preferences. I also consider that an individual allocating a larger amount of resources towards the receiver during the game displays greater prosociality.

The first dictator game was a standard dictator game; in the second game, each player was presented with four pairs of pre-defined allocations. Each player's decisions on their preferred allocations during the second game allow us to classify their type of social preferences. Each game consisted of two rounds, one round with a hypothetical ethnic Roma receiver and one with a hypothetical ethnic Bulgarian receiver. Hence, the ethnic Roma (Bulgarian) receiver is considered an ethnic in-group (out-group) individual for ethnic Roma individuals and an ethnic out-group (in-group) individual for the ethnic Bulgarian individuals. This experimental design allows me to elicit the joint association of ethnic identity and parental preferences with a child's social preferences. Additionally, leveraging the within-subject design, I estimate the prevalence of in-group bias at the individual level and I observe its association between generations. Finally, to explore potential mechanisms of this intergenerational association, I also collected data on expected ethnic discrimination, stereotypes about ethnic minorities, parenting styles, and the frequency of contact with the out-group from each parent in both ethnic groups.

My results indicate that prosocial behavior is prevalent in the sample. The average allocation to the receiver is at least 40% across all groups. I also find that children and ethnic majority individuals are more prosocial than adults and ethnic minorities. However, in-group bias is also widespread. Ethnic Bulgarian children allocate 48.7% of their allocation to the in-group receiver and 44.7% to the out-group receiver. Similarly, ethnic Roma children give 45.7% of their allocation to the in-group receiver and 42.6% to the out-group receiver. In both cases, the difference is not large but it is statistically significant. Among parents, ethnic Roma parents allocate 43.2% to the in-group receiver and 37.4% to the out-group receiver, a statistically significant difference. Conversely, ethnic Bulgarian parents do not show an in-group bias at the group level, as they allocate similar percentages of their allocations

to both receivers. However, I find heterogeneity at the individual level: 16.70% of ethnic Bulgarian parents show an in-group preference, while 11.79% show an out-group preference.

Moreover, I find that prosociality and in-group bias are correlated between generations. On aggregate, the association between parents' and children's prosociality represents 10% of a standard deviation. Relatedly, a one-token increase in parental in-group preference is associated with an 11% increase in children's average in-group preference. However, these estimates differ by ethnic group. My results suggest that prosociality is associated between generations within ethnic Bulgarian households, only when they interact with an out-group receiver, and not within ethnic Roma households, even though the latter estimate is also positive. Lastly, regarding the intergenerational association of in-group preferences, I find that the aggregate estimate is no longer statistically significant once the sample is disaggregated by ethnic group.

Finally, the potential mechanisms that may explain this intergenerational association are different between ethnic groups. In ethnic Bulgarian households, stereotypes are positively correlated with the transmission of prosociality. An increase in parental positive stereotypes towards Roma children is associated with an increase in children's prosociality towards ethnic Roma receivers. On the other hand, daily contact with the ethnic out-group is associated with a weaker intergenerational association of prosociality. None of these mechanisms appear to be relevant in ethnic Roma households. Lastly, regarding the prosocial parenting style, in ethnic Roma households, a prosocial parenting style is associated with an increase in their children's pooled prosociality, but not in ethnic Bulgarian households. Regarding prosociality towards ethnic out-group receivers, a prosocial parenting style is not associated with an increase in children's prosociality in ethnic Bulgarian households.

1.1 Contributions and review of the literature

In this paper, I study the intergenerational association of social preferences and ethnic in-group bias within the same household across different ethnic groups in Bulgaria. I contribute to the literature on social preferences by showing that, besides social preferences, ethnic in-group preference is also positively associated between generations of the same household. Previous literature has shown that ethnic in-group bias in social preferences tends to be prevalent among both adults[1] and children[2]. However, to the best of my knowledge, my study is the first one to show that ethnic in-group bias is also positively correlated between generations. I also find that individual heterogeneity in social preferences explain in-group bias at the group level in both generations. This result relates to a small literature on the relevance of individual heterogeneity in explaining in-group biases in social preferences at the group level (Fong & Luttmer, 2009; Hett et al., 2020; Kranton et al., 2020). My contribution is that I show that this heterogeneity is also present among young children and not only among adults, an aspect also not fully explored in previous studies.

I also contribute to the literature on the intergenerational association of economic preferences by exploring this association in an ethnically diverse setting, contrasting with most of previous studies that have focused on settings that are ethnically very homogeneous. I find suggestive evidence that the

[1]For example, see Ahmed (2010), Bartoš and Levely (2021), Burns (2012), Cetre et al. (2024), Chen et al. (2014), Ferraro and Cummings (2007), Fershtman and Gneezy (2001), Haushofer et al. (2023), Ravetti et al. (2019), Tanaka and Camerer (2016), Van Der Merwe and Burns (2008), and Zhang et al. (2019). Berge et al. (2020) and Bouckaert and Dhaene (2004) fail to find evidence of ethnic in-group bias among adults.

[2]Relevant papers include Angerer and Zhuravleva (2024), Angerer et al. (2016, 2017), Bindra et al. (2020), Elwert et al. (2023), Fehr et al. (2008, 2013), Kudashvili and Lergetporer (2022), and Rao (2019). Note that Friesen et al. (2012) and List et al. (2017) do not find evidence of in-group bias in social preferences among children.

intergenerational association of social preferences appears to be stronger in ethnic majority households than in ethnic minority households. I show that the association is statistically significant only among ethnic majority households and not among ethnic minority ones, even though I cannot rule out that both estimates are equal. Previous studies find that social preferences are also positively correlated between generations (Chowdhury et al., 2022; Doepke & Zilibotti, 2017; Dohmen et al., 2012; Falk et al., 2021; Kosse et al., 2020; Sutter & Untertrifaller, 2020)[3]. The difference between ethnic groups that I find in my setting may indicate that the intergenerational association of social preferences depends, to a larger extent, on the specific context where it is assessed.

Finally, with these findings I also contribute to the literature on discrimination by showing that the emergence of discriminatory behaviours may be correlated with parental behaviours (Angerer et al., 2016; Fehr et al., 2008) and highlighting the relevance of family background in discriminatory behaviours among children (Barron et al., 2023). On a broader scale, I also contribute to the literature on trust and social capital. Strong levels of trust and social capital are correlated with economic growth and economic development (Algan & Cahuc, 2010, 2014; Chetty et al., 2022; Durante et al., 2025). However, my study indicates that inter-group biases may be persistent across generations, and emerge already in early life stages, which should be considered when aiming to promote trust between ethnic groups, and social capital more generally, to foster economic growth and development.

2 Setting: the Roma in Bulgaria

The Roma are the largest ethnic minority group in Europe with approximately 10-12 million individuals (Commission, 2020). In Bulgaria, according to the last census in 2021, they represent 4% of the total population (institute, 2021). However, the Council of Europe estimates that the Roma population in Bulgaria is around 10% (Commission, n.d.). Ethnic Roma individuals have a lower socio-economic status compared to the general population in Bulgaria. According to the Roma Survey 2024 (European Union Agency for Fundamental Rights, 2025), in 2024, 53% of ethnic Roma were at risk of poverty (compared to 22% of the general population) and 35% suffered severe material deprivation (compared to 19% of the general population). This socio-economic gap was also present in areas such as housing, education, or the labor market: 57% of Roma individuals lived in housing deprivation, only 42% completed at least upper-secondary education (compared to 90% of the general population), and rates of active labor market participation (i.e., having a paid job) were 14 percentage points lower among ethnic Roma adults than among the general adult population (European Union Agency for Fundamental Rights, 2025). Importantly, these estimates are not exclusive to the Bulgarian context; similar results are found in other European countries in the same survey (European Union Agency for Fundamental Rights, 2025) or using other survey data (Robayo-Abril & Millan, 2019).

Another characteristic of the Roma population in Bulgaria is the discrimination they suffer due to their ethnic identity. The Roma Survey 2024 (European Union Agency for Fundamental Rights, 2025) also collected data on discriminatory experiences: in 2024, 26% of respondents in Bulgaria indicated that they have felt discriminated against due to their ethnicity when looking for a job during the previous year and 16% indicated that they had felt discriminated against for the same reason while at their workplace. Instances of discrimination towards Roma people are also present in data collected from non-Roma individuals in Bulgaria. According to the Discrimination in the European Union Eurobarometer

[3]Note that Ben-Ner et al. (2017) and Cipriani et al. (2013) do not find evidence of intergenerational association in social preferences in their setting.

in 2023 (Commission, 2024), 43% of Bulgarian citizens indicate that they would feel uncomfortable with a Roma person in the country's highest elected political position (the average share in the European Union is 26%) and 31% indicate that they would not feel comfortable working with a Roma individual (average share in the European Union is 16%). In fact, previous literature shows that anti-Gypsyism is a pervasive issue in European countries that may hinder potential integration efforts towards Roma people (Kende et al., 2021).

Overall, available data indicate that Roma individuals in Bulgaria have a lower socio-economic status than their ethnic majority counterparts and that they are discriminated against due to their ethnic characteristics. These findings help contextualize my study on the determinants of social preferences, as I have implemented a lab-in-the-field experiment in a context with particularly uneven inter-group relations.

3 Data collection and experimental design

3.1 Sampling procedures

I randomly sampled 1,000 households[4] across 121 communities in Bulgaria to survey a parent-child dyad in each of them. 600 of those households were part of 62 low-SES communities with a sizable share of residents from ethnic minorities. These 62 communities were part of a previous randomized controlled trial implemented by some members of the research team. Within each community, the enumerators randomly selected 10 households via a random-walk sampling procedure.[5]

The remaining 400 households are part of a nationally representative sample, stratified by administrative district and urban context. In this case, together with the data collection agency, we followed a two-stage stratified random sampling procedure. First, we stratified the sample on an administrative district level. Then, within each defined stratum, we further stratified the sample by type of settlement according to its population size. Within each of the two strata, we defined the number of households to be sampled according to the distribution of households with children between six and seven years across Bulgaria. Inside every populated stratum, we randomly selected one election district. In the selected district, enumerators conducted a random-walk sampling procedure to select households until they completed the required number of interviews.

I defined the following sample inclusion criteria for the adult and the child in each randomly selected household. Each adult should have been older than 18 years old at the time of the interview, be the main caregiver of the child, and have their main residence in the household. Regarding the child, they should have been born in 2017 or 2018, and they should start primary school from September 2024 onwards[6].

Between April and June of 2024, pairs of trained enumerators visited each of the randomly selected

[4] I originally sampled 1,000 households in 121 communities, but the main body of text reports the results for ethnic Bulgarian and ethnic Roma households only. There are 842 households which are present in 115 communities. I focus on ethnic Bulgarian and ethnic Roma households because the design of the dictator games was aimed at those ethnic groups, as they are the ethnic majority and the largest ethnic minority in the country.

[5] The original sample included 60 communities, but in three of these communities the enumerators did not reach the intended number of households and added three additional communities from the previous RCT.

[6] According to the Bulgarian legislation, all children in the country are supposed to start primary school in September of the year when they turn 7 years old. However, in some cases, pre-school teachers, in accordance with the parents, can suggest that a child starts primary school at an earlier date if the teachers assess that the child has already achieved the required competence level.

households to conduct the interviews. The enumerators established contact with the household members to assess if they met the sample inclusion criteria[7] and to conduct the interviews [8]. After the initial contact, and before starting the procedure, the adults read and signed the informed consent[9] and the child gave the corresponding assent. Both adults and children received compensation for their participation in the interview regardless if it was completed or not: the parents received a voucher to be exchanged for goods in local shops and the children received various stationery items such as colored pencils. Interviews were to be conducted simultaneously in different rooms of the household. If that was not possible, due to a lack of rooms or reluctance by members of the household, enumerators were instructed to indicate interviewees not to interfere with the other interview that was happening at the same time.

3.2 Sample description

Table 1 shows the number of households by ethnic group present in the sample and table 2 presents the main socio-demographic characteristics of parents in both ethnic groups. The main observation is that Roma parents have a lower socioeconomic status than ethnic majority parents. Roma parents have lower educational attainment and lower rates of paid employment than ethnic Bulgarian parents. Specifically, 15% of Roma parents have completed secondary education, compared to 88% of ethnic Bulgarian parents, and 31% have completed any sort of paid work during the previous year, compared to 70% of ethnic Bulgarian parents. There are also differences in household amenities: less than 70% of ethnic Roma households have a latrine or a toilet in their dwelling compared to 94% of ethnic majority households.

Table 3 shows the socio-demographic descriptive statistics of interest for children in both ethnic groups[10]. The socioeconomic divide present in the parents' data also shows up in the children's sample: 47% of the ethnic Roma children have their own room compared to 89% of the ethnic Bulgarian children; 80% of the Roma children have toys in the dwelling compared to 97% of the ethnic Bulgarian households; and 54% of Roma children have books for them in the house compared to 94% of ethnic Bulgarian children. However, the (self-reported) kindergarten attendance is similar across ethnic groups (91% for ethnic Roma children and 96% for ethnic Bulgarian children). Finally, there is a marked difference in Bulgarian language fluency: 47% of ethnic Roma children are fluent in Bulgarian compared to 93% of ethnic Bulgarian children.

[7] In case multiple household members met the inclusion criteria, the selected member was the one first identified by the enumerator to have the required characteristics.

[8] Note that the interview may have not followed directly after the inclusion criteria assessment. Household members may have not been available at that moment to conduct the interview. In those cases, the enumerators arranged a separate date and time to conduct the interview.

[9] This research project received IRB approval from the Ethical Committee of the Faculty of Law, Economics, and Governance at Utrecht University and by BAMOR, the Bulgarian Association of Marketing and Opinion Researchers.

[10] The enumerators did not ask the children to report their preferred ethnicity in the questionnaire. I categorize children's ethnicity according to their parents' self-reported ethnicity.

Table 1: Ethnic groups in the sample

	Count	Percentage	Cumulative Percentage
Ethnic Bulgarians	509	60.45	60.45
Ethnic Roma	333	39.55	100.00
Total	842	100.00	

The table indicates the number of adults in our sample that identified as either ethnic Bulgarian or ethnic Roma. I assume that children's ethnic identity is the same as the ethnic identity reported by the sampled adult in their household.

Table 2: Sample descriptives - Parents

	Ethnic Roma parent	Ethnic Bulgarian parent	Total
Female	0.57	0.62	0.60
	(0.50)	(0.49)	(0.49)
Age	32.65	37.52	35.60
	(9.58)	(9.38)	(9.75)
Completed secondary education	0.15	0.88	0.59
	(0.35)	(0.32)	(0.49)
Employed	0.31	0.70	0.55
	(0.46)	(0.46)	(0.50)
Primary language - Bulgarian	0.78	0.99	0.91
	(0.42)	(0.09)	(0.29)
Primary language with child - Bulgarian	0.75	0.99	0.89
	(0.43)	(0.11)	(0.31)
Latrine/toilet in dwelling	0.62	0.94	0.81
	(0.49)	(0.24)	(0.39)
Running water in dwelling	0.88	0.98	0.94
	(0.32)	(0.12)	(0.23)
Internet connection in dwelling	0.77	0.98	0.90
	(0.42)	(0.15)	(0.30)
Observations	842		

The table reports the average value for each variable of interest by ethnic group of the respondent. Standard errors in parenthesis. All variables except age are dichotomous variables in the 0 - 1 range. All variables are self reported.

Table 3: Sample descriptives - Children

	Ethnic Roma child	Ethnic Bulgarian child	Total
Age	6.32	6.28	6.30
	(0.63)	(0.59)	(0.60)
Female	0.46	0.51	0.49
	(0.50)	(0.50)	(0.50)
Goes to kindergarten/preschool	0.90	0.96	0.94
	(0.30)	(0.19)	(0.25)
Fluent in Bulgarian	0.47	0.93	0.75
	(0.50)	(0.26)	(0.43)
Have own room	0.47	0.89	0.72
	(0.50)	(0.31)	(0.45)
Toys in room	0.80	0.97	0.90
	(0.40)	(0.17)	(0.30)
Books in room	0.54	0.94	0.78
	(0.50)	(0.24)	(0.41)
Observations	842		

The table reports the average values for each variable of interest by ethnic group of the respondent. Standard errors in parenthesis. All variables except age are dichotomous variables in the 0 - 1 range. Fluency in Bulgarian and dwelling assets were assessed by the trained enumerator.

3.3 Instrument design

I developed a survey that included two non-incentivized dictator games and survey questions to elicit the social preferences of all children and parents in both ethnic groups.

In the first dictator game, each player received a hypothetical stake of resources[11] - 20 units of the local currency for the adults and 10 stickers for the children - and were asked to split those items in any way they wanted with a hypothetical receiver. To avoid wealth effects, the game instructions stated that the hypothetical receiver had the same level of material conditions as the sender[12].

I complemented those task-based items with a set of self-reported questions on altruism and trust. There is recent evidence that shows that survey measures can also be a good predictor of choices and decisions in incentivized tasks (Falk et al., 2018, 2023), hence I also include questions developed in the Global Preference Survey and the World Values Survey (Falk et al., 2023; Haerpfer et al., 2022) to elicit social preferences.

[11]The standard procedure uses real payments to incentivize truth-telling, but a meta-analysis of over 120 published dictator games papers by Engel (2011) found that the use of hypothetical questions does not significantly alter the self-reporting of social preferences compared to incentivized decisions.

[12]In Appendix E, I describe the second dictator game and report some descriptive findings.

3.4 Experimental design

All adults and children played both dictator games as the sender. And each player played each game twice: once with a hypothetical ethnic Bulgarian receiver and once with a hypothetical ethnic Roma receiver. Following previous literature (Levitt & List, 2007), I treat the amount allocated to each receiver as a measure of each individual's altruism towards others. I consider altruism to be a prosocial behavior, and I infer that an individual displays social preferences if they allocate any positive amount of their initial resources to the receiver.

Enumerators provided show-cards that represented the stake amounts in the dictator games to all players to help them visualize the amounts they needed to deal with. Additionally, enumerators provided the children with show-cards representing other children to help them visualize that they were interacting with another (hypothetical) player. Appendix A provides a detailed description of the experimental procedures and all the materials used during the surveys.

Finally, to avoid order effects in respondents' answers, I randomized the ethnicity of the first receiver for all the subjects (i.e. some individuals played first with the hypothetical ethnic Bulgarian receiver and others played first with the hypothetical ethnic Roma receiver).

4 Empirical specification

Equation 1 defines a linear model to determine how ethnic identity is associated with both parents' and children's allocations in the standard dictator game[13]. I am interested in the association within each generation; therefore, I estimate the model separately for parents and children.

$$giving_{ir} = \beta_1 ingroup_r + \beta_2 majority_i + \beta_3 majority_i \cdot ingroup_r + Z_i + \varepsilon \tag{1}$$

In equation 1, the dependent variable $giving_i$ represents the standardized amount sent by player i to the receiver in a given round. This amount is standardized across all individuals from the same generation and rounds in the standard dictator game. The independent variable $ingroup_r$ is an indicator variable that equals one for the rounds of the game where the receiver and the sender are from the same ethnic group. Next, $majority_i$ equals one if the sender is ethnic Bulgarian. The parameter β_3 captures the difference in altruism between ethnic Bulgarian and Roma individuals when facing receivers from different ethnic groups. Z is a set of demographic variables (i.e., age and gender of each player), and ε is the error term.

Equation 2 provides a parsimonious model to assess the association between parents' and children's social preferences in our sample:

$$giving_c = \beta_1 giving_p + Z + \varepsilon \tag{2}$$

Here, $giving_c$ is the standardized average amount sent by child c during the dictator game, computed as the mean amount allocated to the two receivers and standardized within the children's sample. The independent variable, $giving_p$, is the corresponding average amount sent by the child's parent, standardized with respect to the parents' sample. Z controls for the age and gender of both the child

[13] All specifications refer to the allocations made by parents and children in the standard dictator game, not their choices in the dictator game using pre-defined allocations.

and their parent.

I also explore differences in the intergenerational association of social preferences between ethnic groups. First, I estimate the following linear model to describe the association of children's prosociality with receivers' ethnicity and their parents' prosociality:

$$giving_{cr} = \beta_1 ingroup_r + \beta_2 giving_{pr} + \beta_3 giving_{pr} \cdot ingroup_r + Z + \varepsilon \tag{3}$$

Here, the dependent variable $giving_{cr}$ is the standardized amount sent by child c in a given round r, standardized across individuals and rounds. The variable $ingroup_r$ is an indicator for whether the hypothetical receiver in round r is an ethnic in-group member. Parental social preferences are captured by the variable $giving_{pr}$, defined as the standardized amount sent by the child's parent to the receiver in the same round r of the game. The parameter β_3 measures whether the transmission of social preferences from parent to child is associated with the ethnicity of the hypothetical receiver. Z controls for the age and gender of the child and parent, and ε is the error term. To capture the association between receiver's ethnicity and the intergenerational association of prosociality within each ethnic group, I estimate equation 3 separately for ethnic Roma and ethnic Bulgarian households.

Finally, the following model describes the association of children's in-group preference[14] with their ethnicity and their parent's in-group preferences:

$$ingpreference_c = \beta_1 majority_c + \beta_2 ingpreference_p + \beta_3 ingpreference_p \cdot majority_c + Z + \varepsilon \tag{4}$$

In this case, the variable $ingpreference_c$ indicates the in-group preference by each child c and variable $ingpreference_p$ indicates the in-group preference by parent p of child c. The variable $majority_c$ is an indicator for whether the child c is ethnic Bulgarian. Therefore, parameter β_1 captures differences in in-group preferences between ethnic groups. Respectively, estimating parameter β_2 provides an estimate of the intergenerational association of in-group preference in our sample. Finally, parameter β_3 provides an estimate of the association between ethnic group and the intergenerational association of in-group preferences.

In all the empirical specifications, I include the age and gender of each player as covariates. The inclusion of other variables potentially related to prosociality, such as educational level or household income, would result in a case of "bad controls" (Angrist & Pischke, 2009)[15]. In my setting, being part of an ethnic minority group may be associated with different levels of prosociality or in-group preference, but it can also be associated with lower educational attainment, a more fragile labor market status, etc., as being an ethnic minority may restrict access to better educational or labor market opportunities (Altonji & Blank, 1999). In fact, there is substantive survey evidence suggesting that this is the case among ethnic Roma individuals in Bulgaria (European Union Agency for Fundamental Rights, 2017, 2025). Therefore, given that one of my variables of interest is ethnic identity and in Bulgaria ethnic identity negatively correlates with other potential determinants of prosocial behavior, such as educational attainment, including these other potential determinants as control variables in

[14] An individual displays an in-group preference if they allocate a larger share of resources to the ethnic in-group receiver than to the ethnic out-group receiver. Given the initial allocations, the variable can take integer values between -20 and 20 in the case of parents, and between -10 and 10 in the case of children. A positive value indicates an in-group preference and a negative value indicates an out-group preference.

[15] As defined by Angrist and Pischke (2009, p. 49): "A second version of the bad control scenario (...) is the inclusion of variables that might partially control for omitted factors, but are themselves affected by the variable of interest."

my specification would result in a case of "bad controls".

However, I include community fixed effects in all specifications[16]. The inclusion of community fixed effects may serve as an indirect proxy for socio-economic status that is not affected by individuals' ethnic identity. Given the high levels of urban segregation reported by Roma individuals in Bulgaria (European Union Agency for Fundamental Rights, 2017)[17], almost all ethnic Roma minority households in our sample come from the same set of communities with similar socio-demographic conditions. Therefore, the inclusion of community fixed effects creates a within-group estimation that soaks up differences in socio-economic status between communities that may explain some of the results.

However, when I estimate differences between ethnic groups in my outcomes of interest, if I include community fixed effects I am leveraging variation between ethnic groups within communities, not between communities. This may limit statistical power on certain estimations and may challenge the external validity of my results, as I am using variation between ethnic groups in a sub-sample of communities. Specifically, as displayed in figure D.7 in the appendix, there are only 37 out of 115 communities with both ethnic Bulgarian and ethnic Roma households[18]. Notwithstanding that limitation, I argue that the inclusion of community fixed effects is more beneficial than not due to their role in controlling for socio-economic differences between communities, which can play a substantial role in the emergence and development of social preferences (Falk et al., 2021).

5 Results

5.1 Main results

5.1.1 Influence of ethnic identity

Prosocial behavior is widespread in our sample. Figure 1 displays the average amount allocated to receivers by ethnic group and generation, and we observe that at least 40% of the initial allocation is allocated to receivers across all groups. However, these allocations are not equal between generations and ethnic groups.

Between generations, children are more prosocial than their parents. Ethnic Roma children share, on average, 44.1% of their allocation with receivers, while their parents allocate 40.3%. Similarly, ethnic Bulgarian children share 46.7% of their resources with receivers, while their parents share 43.3%. In both cases, the difference between allocations is statistically significant. Between ethnic groups, ethnic majorities allocate a larger amount to receivers than ethnic minorities. The average amount shared with receivers in ethnic Roma households represents 42.2% of their initial allocation. In ethnic Bulgarian households, the average allocation is 45% of their initial endowment.

The estimates in figure 1 represent average allocations across receivers. While informative, average allocations may mask differences in prosocial behavior depending on receiver's ethnicity because, as described extensively in the scientific literature (Balliet et al., 2014; Lane, 2016), individuals are ex-

[16] Additionally, I cluster standard errors at the community level to account for potential intra-cluster correlation between the individuals in each community.

[17] According to the 2016 Survey on Minorities and Discrimination in EU, 83% of Roma individuals in Bulgaria report that all (or most) of their neighbors are also Roma.

[18] As a robustness test, in appendix C, I replicate all the main empirical specifications without community fixed effects. In general, all estimates remain qualitatively the same in size and statistical significance. The only difference arises with respect to the intergenerational transmission of in-group preferences. In that case, I find evidence of average transmission at the 10% level - not 5%. Moreover, the intergenerational association within ethnic Roma households is statistically significant, while it is not significant within ethnic Bulgarian households.

pected to be more prosocial towards receivers from their same ethnic group than from a different ethnic group. To explore these potential differences in prosociality, figures 2 and 3 display the amount shared by children and parents with in-group and out-group receivers respectively. Additionally, figures D.1 and D.2 display the distribution of parental and children's giving by ethnic group and receiver. Lastly, tables 4 and 5 present the estimates of the linear model defined in equation 1 for children and parents in our sample.

I find that parental and children's prosocial behavior is associated with their ethnic identity, even though the association is not statistically significant among ethnic Bulgarian parents. In figure 2, I show that ethnic Roma (ethnic Bulgarian) children give 45.7% (48.7%) of their allocation to the in-group receiver and 42.6% (44.7%) to the out-group receiver[19]. In a similar manner, as shown in figure 3, ethnic Roma parents share 43.2% of their allocation with the in-group receiver and 37.4% with their out-group receiver. Ethnic Bulgarian parents, on the other hand, do not display an in-group preference in their prosociality. Figure 3 indicates that they share approximately 43% of their allocation with both in-group and out-group receivers, without displaying a statistically significant in-group preference.

The linear model estimates confirm the results displayed in the figures. Table 4 shows that, on average, interacting with an in-group receiver is associated with an increase of 16% of a standard deviation in children's prosociality compared to interacting with an out-group receiver. Exploring this association by children's ethnicity, I find that for ethnic Roma children, interacting with an in-group receiver is associated with an increase of 0.14 standard deviations in their prosociality, while for ethnic Bulgarian children it is associated with an increase of 0.18 standard deviations. Taken together, these results indicate that children in our sample exhibit an in-group preference in their social preferences. The estimates for the parents' sample, displayed in table 5, indicate that ethnic Roma parents show a difference of 0.23 standard deviations between their allocations, showing that ethnic Roma parents also display an in-group bias in their social preferences. Finally, for ethnic Bulgarian parents, the estimates presented in table 5 confirm the lack of a statistically significant in-group preference. However, those group-level estimates mask heterogeneity at the individual level.

[19] Also, as found previously in the literature (Angerer et al., 2016; Fehr et al., 2008) and in figure D.4, children become more prosocial as they age. However, note that in my case, I do not find a monotonic increase of in-group preferences over time.

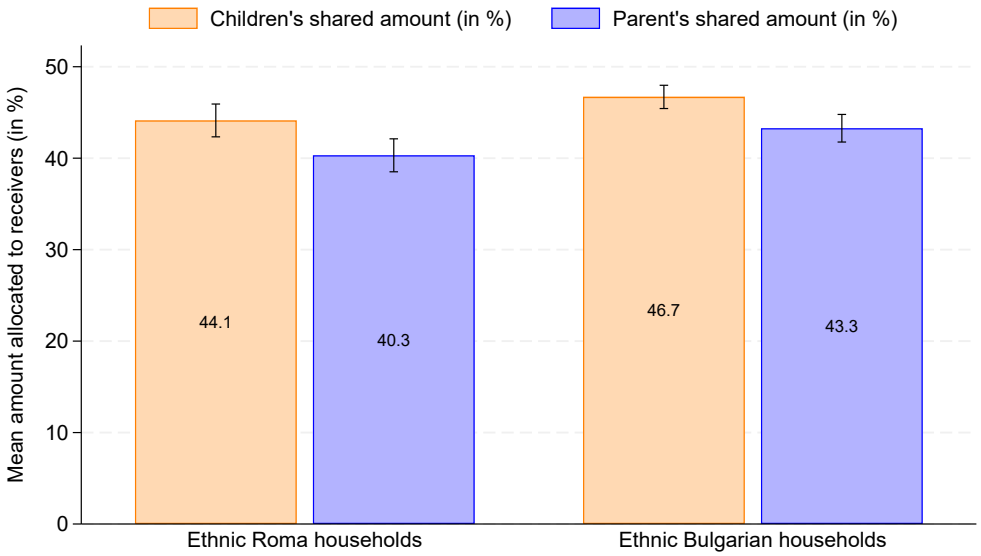


Figure 1: Shared amount between receivers by ethnic group and generations

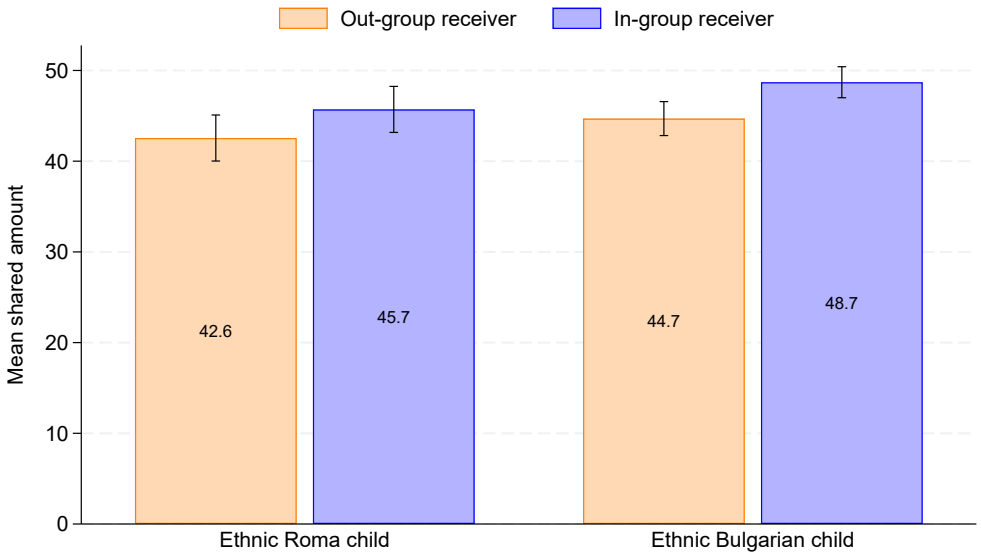


Figure 2: Average amount shared by children (in %)

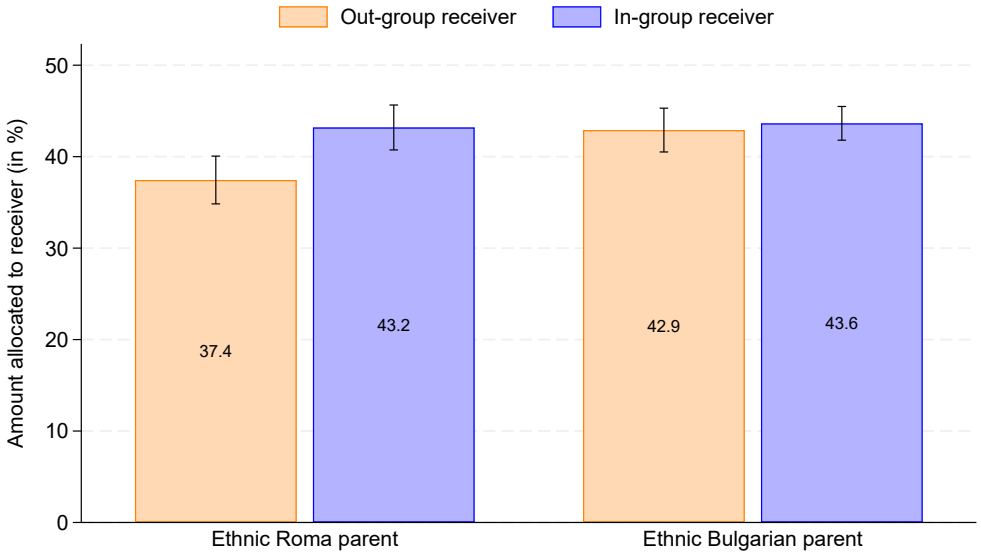


Figure 3: Average amount shared by parents (in %)

Table 4: Ethnic identity influence on children's social preferences

	Child's prosociality	Child's prosociality	Child's prosociality	Child's prosociality
In-group receiver	0.168***	0.168***	0.144**	0.144**
	(0.038)	(0.038)	(0.060)	(0.060)
Ethnic Bulgarian sender		0.126	0.107	0.108
		(0.109)	(0.109)	(0.109)
Ethnic Bulgarian sender × In-group receiver			0.039	0.039
			(0.075)	(0.075)
Constant	-0.084***	-0.160**	-0.148**	-0.161
	(0.019)	(0.065)	(0.065)	(0.118)
Children's demographic variables	No	No	No	Yes
Observations	1684	1684	1684	1684
Receiver's in-group interaction - point estimate			.183	.183
Receiver's in-group interaction - p-value			.000152	.000154

The table presents the linear regression estimates of children's prosociality on receiver and sender's ethnicity. The dependent variable indicates the amount, standardized across individuals and rounds, shared by each child with the receiver during each round of the game. Ethnic Bulgarian sender is a bivariate variable that equals 1 if the sender identifies as ethnic Bulgarian and equals 0 if the sender identifies as ethnic Roma. In-group receiver is a bivariate variable that equals 1 if the game's receiver is from the same ethnic group as the sender, and equals 0 otherwise. Receiver's in-group interaction tests the null hypothesis that the partial association of increased children's prosociality when interacting with an in-group receiver equals zero for ethnic Bulgarian senders. Standard errors clustered at the community level. We control for age and gender of each player. We include community fixed effects. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 5: Ethnic identity influence on parental social preferences

	Parental prosociality	Parental prosociality	Parental prosociality	Parental prosociality
In-group receiver	0.111***	0.111***	0.233***	0.233***
	(0.039)	(0.039)	(0.043)	(0.043)
Ethnic Bulgarian sender		0.249**	0.350***	0.356***
		(0.105)	(0.118)	(0.117)
Ethnic Bulgarian sender × In-group receiver			-0.203***	-0.203***
			(0.069)	(0.069)
Constant	-0.055***	-0.206***	-0.267***	-0.082
	(0.019)	(0.068)	(0.072)	(0.147)
Parents' demographic variables	No	No	No	Yes
Observations	1683	1683	1683	1683
Receiver's in-group interaction - point estimate			.0304	.0304
Receiver's in-group interaction - p-value			.569	.569

The table presents the linear regression estimates of parental prosociality on receiver and sender's ethnicity. The dependent variable indicates the amount, standardized across individuals and rounds, shared by each parent with the receiver during each round of the game. Ethnic Bulgarian sender is a bivariate variable that equals 1 if the sender identifies as ethnic Bulgarian and equals 0 if the sender identifies as ethnic Roma. In-group receiver is a bivariate variable that equals 1 if the game's receiver is from the same ethnic group as the sender, and equals 0 otherwise. Receiver's in-group interaction tests the null hypothesis that the partial association of increased parental prosociality when interacting with an in-group receiver equals zero for ethnic Bulgarian senders. Standard errors clustered at the community level. We control for age and gender of each player. We include community fixed effects. Stars indicate statistical significance: * 10% ** 5% *** 1%

In table 6, we present the share of parents in both ethnic groups that display an in-group preference or an out-group preference in their allocations[20]. An important finding is that a majority of parents and children do not show an in-group or out-group preference, which may be striking given the group-level estimates discussed in previous tables. However, we find that 16.70% of ethnic Bulgarian parents present a relative in-group preference, while 11.79% show a relative out-group preference. For reference, 23.42% of ethnic Roma parents present an in-group preference, while only 4.8% display an out-group preference. Among children, table 7 shows that approximately 28% of ethnic Roma and 23.38% of ethnic Bulgarian children display an in-group preference, and approximately 12% of children from both ethnic groups display an out-group preference. Therefore, while at the group level ethnic Bulgarian parents do not display an in-group bias, there are differences at the individual level, as similar shares of parents display an in-group and an out-group preference that cancel out when estimating the average allocation at the group level.

We complement the task-based estimates for the parents' sample with the survey-based results presented in table D.1. The survey responses indicate that prosocial behaviors are widespread across ethnic groups as a hypothetical donation to charity and their willingness to donate are positive for both ethnic groups. The responses also highlight the relatively higher prosociality of ethnic Bulgarian parents that we observe in figure 3: they would give more to charity and be more willing to do so than ethnic Roma parents. Regarding their trust levels towards other groups, ethnic Bulgarian parents have higher trust towards others in general - 26% of ethnic Bulgarian parents indicate that most people can be trusted compared to 13% of ethnic Roma parents. However, ethnic Bulgarian parents have a lower trust towards members of other ethnic groups than ethnic Roma parents.

Taken together, these results indicate that prosocial behavior is widespread, but almost all groups exhibit an in-group bias in their social preferences. However, this in-group bias at the group level appears associated with the behavior of a relatively small share of individuals within each group. These findings are aligned with previous literature. Several studies have documented that relative preference for the in-group is common across human groups - for both adults (Cetre et al., 2024; Fershtman & Gneezy, 2001) and children (Fehr et al., 2008, 2013) - and that in-group preferences at the group level are driven by differences at the individual level (Fong & Luttmer, 2009; Hett et al., 2020; Kranton et al., 2020).

[20]We define that a sender displays an in-group preference if the amount allocated to the in-group receiver is larger than the amount allocated to the out-group receiver. Relatedly, a sender displays an out-group preference if the amount allocated to the out-group receiver is larger than the amount allocated to the in-group receiver.

Table 6: Parental in-group preference at the individual level

	Ethnic Bulgarian parents	Ethnic Roma parents	Total
Preference for the in-group	16.70% (85)	23.42% (78)	19.36% (163)
No in-/out-group preference	71.51% (364)	71.77% (239)	71.62% (603)
Preference for the out-group	11.79% (60)	4.80% (16)	9.03% (76)
Total	100.00% (509)	100.00% (333)	100.00% (842)

The table presents the number and percentage of ethnic Bulgarian and Roma parents that display a preference for the in-group or the out-group and those that do not display a specific group preference in their prosociality.

Table 7: Children's in-group preference at the individual level

	Ethnic Bulgarian children	Ethnic Roma children	Total
Preference for the in-group	23.38% (119)	27.93% (93)	25.18% (212)
No in-/out-group preference	64.64% (329)	59.46% (198)	62.59% (527)
Preference for the out-group	11.98% (61)	12.61% (42)	12.23% (103)
Total	100.00% (509)	100.00% (333)	100.00% (842)

The table presents the number and percentage of ethnic Bulgarian and Roma children that display a preference for the in-group or the out-group and those that do not display a specific group preference in their prosociality.

5.1.2 Intergenerational transmission of social preferences

Prosociality and in-group preferences are transmitted between generations. Figure 4 displays a binned scatterplot of the average amount shared by parents and children in our sample. The plotted linear fit across the data points presents a positive slope, providing suggestive evidence that prosociality is associated between generations of the same household.

The estimates in table 8 confirm that intuition: an increase of one standard deviation in parental prosociality is associated with 11% of a standard deviation increase in their children's prosociality[21]. And, while 0.11 standard deviations may not seem like a large estimate, it is aligned with results found in previous literature. For example, Kosse et al. (2020) and Dohmen et al. (2012) estimate the correlation coefficient between parental and children's prosociality to be between 0.14 and 0.25 standard deviations. Similarly, Sutter and Untertrifaller (2020) find that children are 17% more likely to cooperate in a prisoners' dilemma game if their parents cooperate as well, and Chowdhury et al. (2022) estimate a positive and statistically significant association between parental and children's social preferences in a sample of 500 families in Bangladesh.

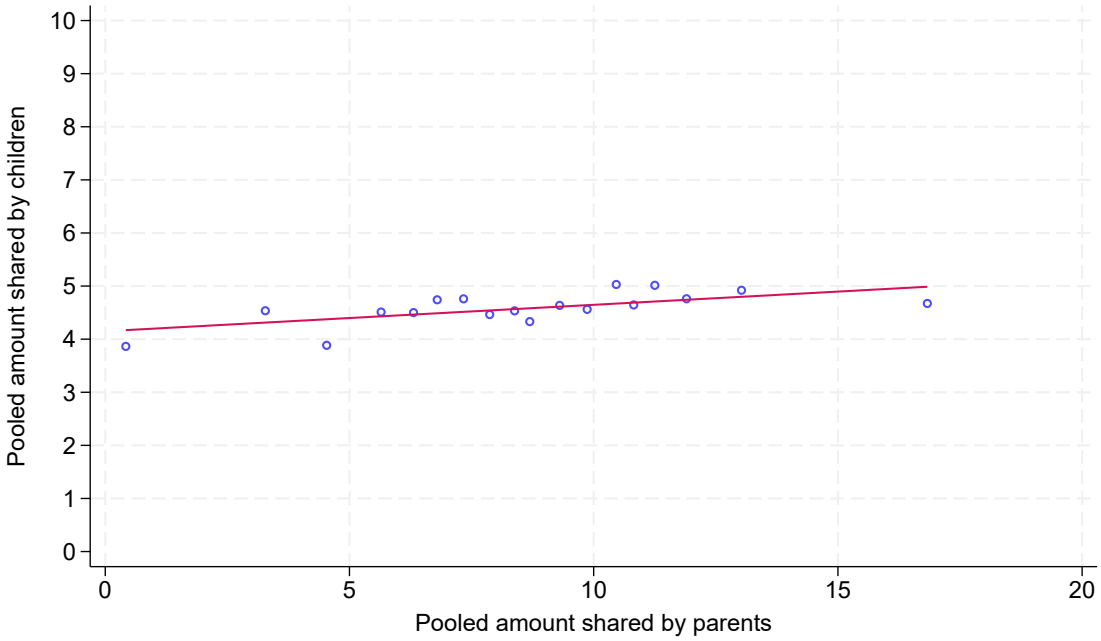


Figure 4: Intergenerational transmission of prosociality

However, we find that the intergenerational transmission of prosociality varies between ethnic groups. In ethnic Roma households, increases in parental prosociality are not associated with changes in children's prosociality. As shown in table 9, the estimate for intergenerational association in ethnic Roma households is positive and close to the coefficient for the whole sample (0.09 standard deviations), but it is not statistically significant. Conversely, in ethnic Bulgarian households, parental and children's prosociality are positively associated, but only when they interact with an out-group receiver.

[21]We also assess if the intergenerational transmission is stronger at different points of the distribution of prosociality. The estimates indicate that parents with relatively higher levels of prosociality transmit them to their children. The estimated probabilities by the linear probability model, in table D.2, indicate that having a parent above the median prosocial behavior is associated with an increase of 15 percentage points in the probability that their child's prosocial behavior is also above the median. Respectively, the estimates in table D.3 show that children with a parental prosociality level above the median have a prosociality level 0.23 standard deviations higher than children with parental prosociality below the median.

Table 8: Intergenerational transmission of social preferences

	Child's prosociality (pooled)	Child's prosociality (pooled)
Parental prosociality (pooled)	0.113**	0.112**
	(0.051)	(0.051)
Constant	-0.003***	-0.360
	(0.000)	(0.528)
Children's demographic variables	No	Yes
Parents' demographic variables	No	Yes
Observations	826	826

The table presents linear regression estimates of the intergenerational association between parents' and children's prosociality. We pool the amount shared by each child and its parents across their respective rounds of the dictator game and then standardize with respect to their samples. The dependent variable indicates the average amount across rounds shared by each child during the dictator game, and the independent variable is the average amount across rounds shared by their parent. Both variables are standardized with respect to their samples. Standard errors are clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

As shown in the second column of table 9, the intergenerational transmission coefficient equals 13% of a standard deviation when both household members interact with the out-group receiver. When they both interact with an in-group receiver, the coefficient shrinks to 3% of a standard deviation and it becomes indistinguishable from zero.

Regarding the transmission of in-group preferences, the first two columns in table 10 show that a one-token increase in parental in-group preference is associated with a 0.042 token increase in children's in-group preference – an approximately 11% increase with respect to children's average in-group preference. However, in this case, the transmission of in-group preferences is not different between ethnic groups. In both ethnic groups, as displayed in table 10, we cannot reject the null hypothesis of no intergenerational transmission of in-group preferences. Lastly, I cannot contextualize these estimates with previous literature. To the best of my knowledge, there are no published studies that estimate a similar metric that would allow me to assess its economic relevance.

To the best of our knowledge, the only other paper that studies the intergenerational transmission of social preferences between different ethnic groups is Barron et al. (2023). In their paper, the authors study the prevalence of out-group discrimination between Jordanian host and Syrian refugee children in Jordan using several experimental tasks. Their main results are the high levels of generosity among children in both ethnic groups and low levels of discrimination between ethnic groups Barron et al. (2023). Regarding the former, they observe relatively large shares of children, between 33 and 45% depending on the sample, giving more resources to the receiver than themselves during the experimental tasks, even when the receiver is part of the ethnic out-group.

One of their main contributions is that they can link children's prosocial behaviors to parental and family narratives. They find that Jordanian children from families with a refugee background do not engage in discriminatory behavior against the Syrian refugee children, and that parental opinions about the Syrian refugee crisis are correlated with their children's degree of out-group discrimination. With those findings, the authors highlight the importance of family narratives in shapeing children's economic preferences and behaviors.

Table 9: Intergenerational transmission of prosociality within ethnic groups

	Child's prosociality	Child's prosociality
In-group receiver	0.114*	0.185***
	(0.060)	(0.049)
Parental prosociality	0.092	0.142***
	(0.084)	(0.048)
In-group receiver × Parental prosociality	0.024	-0.104**
	(0.067)	(0.049)
Constant	-0.099	0.246
	(0.718)	(0.600)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1,018
Joint interaction test - estimate	.116	.0379
Joint interaction test - p-value	.251	.602
Equality of coefficients	.595	.595

The table presents the linear regression estimates of child's prosociality on parental prosociality and receiver's ethnicity. First column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game on which the receiver is from the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household to the receiver during each round of the game. The joint interaction tests, separately for each ethnic group, the null hypothesis of null intergenerational association when interacting with an in-group receiver. The equality of coefficients row reports the p-value for the null hypothesis of an equal intergenerational association estimate between ethnic groups. Standard errors are clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 10: Intergenerational transmission of in-group preferences

	Child's in-group preference	Child's in-group preference	Child's in-group preference	Child's in-group preference
Parental in-group preference	0.040**	0.042**	0.063	0.063
	(0.018)	(0.018)	(0.038)	(0.038)
Ethnic Bulgarian household			-0.029	-0.011
			(0.305)	(0.311)
Ethnic Bulgarian household × Parental in-group preference			-0.031	-0.028
			(0.047)	(0.047)
Constant	0.329***	0.695	0.337*	0.668
	(0.010)	(1.032)	(0.185)	(1.068)
Children's demographic variables	No	Yes	No	Yes
Parents' demographic variables	No	Yes	No	Yes
Observations	825	825	825	825
Sender's ethnicity interaction - point estimate			.032	.0349
Sender's ethnicity interaction - p-value			.159	.134

The table presents the linear regression estimates of the intergenerational association between parental and children's in-group preferences. The dependent variable is the child's in-group preference - the difference between the amount allocated to the in-group receiver and the out-group receiver during the dictator game - and the main independent variable is parental in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. The variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The joint interaction tests the null hypothesis of a null intergenerational association of in-group preferences in ethnic Bulgarian households. Standard errors clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

In general, my paper and Barron et al. (2023) provide evidence that children's social preferences are jointly associated with their parents' behavior and ethnic identity. However, I argue that my experimental design provides a more precise measure of the intergenerational transmission of in-group preferences. In my study, I elicit social preferences directly from both parents and children, which provides a comparable measure of prosocial behavior for both generations within the same household. Conversely, Barron and coauthors elicit children's prosocial behavior and estimate its association with parental attitudes, opinions, and family history. Their approach provides a novel angle in cultural transmission, but it does not provide a comparable measure of social preferences for both parents and children.

5.2 Mechanisms

In this section, I assess different parental behaviors and attitudes[22] as mechanisms to explain children's level of prosociality, the intergenerational association of prosociality, and the intergenerational association of in-group preferences. Regarding the first element, I present results related to the association of these parental characteristics with (i) children's average prosociality towards receivers, and with (ii) children's prosociality towards out-group receivers specifically.

5.2.1 Parenting styles

Parenting style is defined as the degree of control and response by parents over their child's actions (Baumrind, 1967). In economics, parenting styles are understood as a mechanism of preference transmission and as an input in children's human capital production function (Cobb-Clark et al., 2019; Doepke & Zilibotti, 2017). Specifically, parents are seen as economic agents that combine Beckerian altruism and paternalism (Doepke & Zilibotti, 2017). Parents care about the welfare and utility maximization of their children (i.e. Beckerian altruism), but they have defined preferences on how this utility maximization should happen (i.e. paternalism) (Doepke & Zilibotti, 2017).

In our case, we identify parents with a prosocial parenting style by including an item of the World Values Survey that lists qualities that children can learn at home in the parental survey[23]. We define parents with a prosocial parenting style as those that chose 'Respect for other people' and/or 'Unselfishness' among their preferred qualities that children can learn at home. We assume that parents with a prosocial parenting style will strengthen the prosocial behavior of their children more than parents that do not indicate a prosocial parenting style.

Our estimates suggest that a prosocial parenting style is associated with the intergenerational association of social preferences, but with differences between ethnic groups[24]. The estimates in table 11 indicate a positive, but non-statistically significant, association with children's average prosociality. Once we assess differences by ethnic group, the intergenerational association is positive and statistically significant at the 10% level for ethnic Roma households[25], while remaining close to zero in ethnic Bulgarian households. Regarding the association between a prosocial parenting style and child's prosociality towards the out-group receiver, the estimates in table 12 are not statistically different from zero

[22] I collected data on parenting styles, parental expectations of discrimination towards their children, inter-group ethnic stereotypes and the frequency of contact with the out-group for all the parents in the sample.

[23] The complete list of qualities is the following: good manners, independence, hard work, feeling of responsibility, imagination, respecting parents, respect for other people, thrift saving money and things, determination and perseverance, religious faith, unselfishness, obedience. Respondents are asked to pick the five they found especially important.

[24] 75% of the parents in both ethnic groups indicate a prosocial parenting style - as displayed in figure D.5.

[25] This association remains statistically significant after controlling for parental prosociality.

in either ethnic group.

I also observe that, regarding the intergenerational transmission of prosociality, a prosocial parenting style is associated with a weaker transmission of prosociality in ethnic Bulgarian households towards Roma receivers in table D.5. In terms of the intergenerational association of in-group preferences, in table 13, the estimates are positive and significant among parents without a prosocial parenting style, but close to zero in parents with prosocial parenting style. Assessing this transmission between ethnic groups, as displayed in table D.8, I find that the estimates are indistinguishable from zero for all parents in both ethnic groups.

Taken together, these results indicate that a prosocial parenting style is associated with a less group-biased prosociality among children. This association is aligned with the nature of the values upheld by parents with a prosocial parenting style: unselfishness and respect for others may be correlated with a more "universalist" worldview (Enke, 2024) that overcomes group differences.

Table 11: Parenting styles and child's prosociality

	Child's prosociality (pooled)	Child's prosociality (pooled)	Child's prosociality (pooled)
Prosocial parenting style	0.133	0.277*	0.267*
	(0.103)	(0.159)	(0.156)
Ethnic Bulgarian child		0.338*	0.300
		(0.194)	(0.196)
Prosocial parenting style × Ethnic Bulgarian child		-0.246	-0.236
		(0.211)	(0.207)
Parental prosociality (pooled)			0.106*
			(0.055)
Constant	0.393	0.260	0.150
	(0.631)	(0.656)	(0.642)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Sample			
Observations	842	842	842
Prosocial parenting and ethnic Bulgarian households - estimate		.0309	.0303
Prosocial parenting and ethnic Bulgarian households - p-value		.817	.816

The table presents the estimations of a linear regression of the average amount shared in the dictator game by the children on parents' prosocial parenting style. The dependent variable indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each child during the game. Parental prosociality indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each parent during the game. Prosocial parenting style is a bivariate variable that equals 1 for those children with parents that present a prosocial parenting style. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 12: Parenting styles and child's prosociality towards the out-group

	Child's prosociality	Child's prosociality	Child's prosociality
Prosocial parenting style	0.068	0.239	0.225
	(0.110)	(0.160)	(0.156)
Ethnic Bulgarian child		0.372*	0.324
		(0.211)	(0.211)
Prosocial parenting style × Ethnic Bulgarian child		-0.291	-0.279
		(0.225)	(0.219)
Parental prosociality			0.113**
			(0.044)
Constant	-0.037	-0.193	-0.285
	(0.576)	(0.610)	(0.593)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Sample			
Observations	842	842	842
Prosocial parenting and ethnic Bulgarian households - estimate		-.0526	-.0547
Prosocial parenting and ethnic Bulgarian households - p-value		.73	.712

The table presents the estimations of a linear regression of the amount shared in the dictator game by the children on parents' prosocial parenting style. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the out-group receiver during the game. Parental prosociality indicates the average amount, standardized with respect to the whole sample, shared by each parent with the out-group receiver during the game. Prosocial parenting style is a bivariate variable that equals 1 for those children with parents that present a prosocial parenting style. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 13: Parenting styles and intergenerational transmission of in-group preferences

	Children's in-group preference
Parental in-group preference	0.062**
	(0.028)
Prosocial parenting style	0.270
	(0.174)
Prosocial parenting style × Parental in-group preference	-0.029
	(0.037)
Constant	0.546
	(1.043)
Children's demographic variables	Yes
Parents' demographic variables	Yes
Observations	825
Intergen. assoc. of in-group preference among prosoc. parents - estimate	.0322
Intergen. assoc. of in-group preference among prosoc. parents - p-value	.178

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Prosocial parenting style is an indicator variable that equals 1 for parents that indicate a prosocial parenting style and 0 for parents that do not. The variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypotheses of no intergenerational association of parental in-group preference among prosocial parents across the whole sample and the different ethnic groups. Standard errors clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

5.2.2 Frequency of contact with the out-group

Inter-group contact theory (Allport, 1954) predicts that contact with an out-group, under certain conditions, should reduce group prejudice. These conditions are the following: equal status between groups, common goals, intergroup cooperation, and support from the authorities (Pettigrew, 1998). According to Pettigrew (1998), contact under these conditions reduces prejudice because they provide a setting for the involved individuals to learn about the out-group, correcting potential negative views and reappraising them, while changing their behavior towards them and generating friendship ties between the groups. A recent meta-analysis of pre-registered inter-group interventions supports these hypotheses: contact with the out-group reduces prejudice by 0.08 standard deviations, albeit the Allport conditions do not appear necessary to generate this positive effect (Lowe, 2025).

In this study, I did not implement an inter-group contact intervention, but I collected a measure of interaction between the groups to assess if it was associated with the transmission and emergence of social preferences. Specifically, each parent assessed their frequency of interaction with an ethnic out-group member on a Likert scale from 1 to 7, with options ranging from the least frequent (i.e., "Never") to the most frequent (i.e., "Daily") option[26]. I find that ethnic Roma parents interact much more frequently with ethnic Bulgarian individuals than ethnic Bulgarian parents interact with ethnic Roma individuals. Figure D.6 shows that around 75% of ethnic Roma parents interact daily with ethnic Bulgarians, compared to 36% of ethnic Bulgarians that interact daily with Roma individuals. This result is consistent with the high levels of urban segregation reported by Roma individuals and the relatively smaller number of Roma individuals in Bulgaria, as these two elements make it more probable that an ethnic minority individual interacts with an ethnic Bulgarian than the other way around.

Table 14: Parental contact with the outgroup and child's prosociality

	Child's prosociality (pooled)	Child's prosociality (pooled)	Child's prosociality (pooled)
Daily contact with the out-group	0.072	0.197	0.195
	(0.093)	(0.185)	(0.185)
Ethnic Bulgarian child		0.304	0.272
		(0.217)	(0.217)
Daily contact with the out-group × Ethnic Bulgarian child		-0.186	-0.185
		(0.221)	(0.220)
Parental prosociality (pooled)			0.109*
			(0.056)
Constant	0.440	0.317	0.199
	(0.647)	(0.650)	(0.639)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	842	842	842
Daily contact with the outgroup and ethnic Bulgarian households - estimate		.0112	.00955
Daily contact with the outgroup and ethnic Bulgarian households - p-value		.917	.928

The table presents the estimations of a linear regression of the average amount shared in the dictator game by the children on parents' daily contact with the out-group. The dependent variable indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each child during the game. Parental prosociality indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each parent during the game. Daily contact with the out-group is a bivariate variable that equals 1 for those children's parents that have daily contact with the ethnic out-group. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

[26]For ethnic Bulgarian parents, the question referred to ethnic Roma individuals exclusively, and for Roma parents it referred to ethnic Bulgarian individuals exclusively.

Table 16: Parental contact with the outgroup and intergenerational transmission of in-group preference

	Children's in-group preference
Parental in-group preference	0.062***
	(0.019)
Daily contact with the out-group	0.057
	(0.209)
Daily contact with the out-group × Parental in-group preference	-0.050*
	(0.027)
Constant	0.799
	(1.023)
Children's demographic variables	Yes
Parents' demographic variables	Yes
Observations	825
Intergen. assoc. of in-group preference among parents with daily outgroup contact - estimate	.0122
Intergen. assoc. of in-group preference among parents with daily outgroup contact - p-value	.639

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Daily contact with the out-group is an indicator variable that equals 1 for parents that have daily contact with the out-group and 0 for parents not in daily contact with the out-group. The variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interaction tests the null hypothesis of no intergenerational association of parental in-group preference among parents with daily contact with their ethnic out-group across the whole sample and the different ethnic groups. Standard errors clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 15: Parental contact with the outgroup and child's prosociality towards the out-group

	Child's prosociality	Child's prosociality	Child's prosociality
Daily contact with the out-group	0.071	0.077	0.070
	(0.103)	(0.187)	(0.190)
Ethnic Bulgarian child		0.152	0.121
		(0.229)	(0.233)
Daily contact with the out-group × Ethnic Bulgarian child		0.002	-0.009
		(0.231)	(0.231)
Parental prosociality			0.114**
			(0.045)
Constant	-0.047	-0.063	-0.160
	(0.618)	(0.609)	(0.592)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	842	842	842
Daily contact with the outgroup and ethnic Bulgarian households - estimate		.0788	.061
Daily contact with the outgroup and ethnic Bulgarian households - p-value		.532	.618

The table presents the estimations of a linear regression of the amount shared in the dictator game by the children on parents' daily contact with the out-group. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the out-group receiver during the game. Parental prosociality indicates the average amount, standardized with respect to the whole sample, shared by each parent with the out-group receiver during the game. Daily contact with the out-group is a bivariate variable that equals 1 for those children's parents that have daily contact with the ethnic out-group. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Broadly, the frequency of inter-group contact does not explain the prosocial behavior of children in our sample. All the estimates in table 14 and table 15 are statistically equal to zero, for both ethnic Roma and ethnic Bulgarian households. Moreover, these estimates do not change if I control for parental prosociality.

However, in table D.6, when I include the frequency of inter-group contact I find differences in the intergenerational association of prosociality between ethnic groups. I find that, in ethnic Bulgarian households, parental daily contact with the ethnic out-group is associated with no intergenerational

transmission of prosociality towards Roma receivers. For those parents, the association estimate is equal to 0.08 standard deviations (p-value > 0.10), which contrasts with the estimate of 0.17 standard deviations (p-value < 0.01) for ethnic Bulgarian parents without daily contact with the out-group. On the other hand, for ethnic Roma households, the inclusion of daily contact with the out-group in the empirical model does not change the estimates for the intergenerational association of prosociality.

Regarding the intergenerational transmission of in-group preferences - estimates shown in table 16 - I find that in-group preferences are not associated between generations in those households that are in daily contact with the out-group. Assessing differences between ethnic groups[27], I also find no evidence of in-group preference transmission in either ethnic group.

Therefore, in this setting, parental contact with the ethnic out group does not explain children's prosocial behavior and it is associated with (i) weaker intergenerational association of prosociality towards the out-group in ethnic Bulgarian households and with (ii) no intergenerational transmission of in-group preferences.

5.2.3 Stereotypes

Our beliefs and perceptions about others drive our behavior towards them. Alesina et al. (2023) show that respondents in a large-scale survey implemented in the United States, the United Kingdom, Italy, Sweden, Germany and France hold misperceived beliefs about the number and composition of immigrants in their countries. On average, respondents overestimate the number of total migrants in the country — the average misperception between the perceived and the actual number is at least 15 percentage points in all countries — and tend to think that migrants are much more culturally distant than what they actually are. Crucially, those perceptions translate into actual behavior. When survey respondents are nudged to think about immigration during the survey, they displayed lower support for redistribution and lower donations to charity (Alesina et al., 2023).

On a smaller scale, Tanaka and Camerer (2016) highlight that prosocial behaviors between Vietnamese, Khmer, and Chinese ethnic groups in Vietnam are explained by inter-group perceived stereotypes. They find that ethnic Khmers are perceived as warm individuals, but with low status and low competence by the other two ethnic groups (Tanaka & Camerer, 2016). Those perceptions lead Vietnamese and Chinese individuals to display out-group bias towards the Khmer in experimental games that involved trust but not in experimental games that involved help and altruism[28] (Tanaka & Camerer, 2016).

Given the previous examples, I hypothesize that perceptions about the ethnic out-group may be associated with children's prosocial behavior and the intergenerational association of social preferences in our context. Parents that hold more positive stereotypes about the ethnic out-group may transmit those to their children, prompting higher levels of prosociality among children towards the out-group receiver.

To test the association between parental stereotypes and children's prosocial behavior, I elicited the perceptions about children in both ethnic groups from all the parents in our sample. Concretely, all parents provided their level of agreement with different statements[29] regarding ethnic Bulgarian

[27] See table D.9

[28] For a recent and more extensive review on the literature about perceptions and beliefs about others, especially about members of an out-group, see Bursztyn and Yang (2022).

[29] The writing of the statements is as follows: "In general, do you think that Bulgarian/Roma majority children: have a

and ethnic Roma children on a scale from zero to ten. Then, I standardize and aggregate all their individual ratings across statements to create an index variable. The index variable represents the degree of stereotypical views hold by each parent. I estimate two indexes: one for stereotypes towards ethnic Bulgarian children and one for stereotypes towards ethnic Roma children. Variables are coded such that a higher score in the index represents more positive stereotypes.

Figure 5 shows that parents from both ethnic groups tend to have more negative views towards the out-group children than their in-group children. Hence, ethnic Bulgarian parents have more negative views of Roma children than of ethnic Bulgarian children, and Roma parents have more negative views of ethnic Bulgarian children than of ethnic Roma children. However, the shape of the distributions suggests that ethnic Bulgarian parents hold, on average, more negative views about children from their out-group than ethnic Roma parents.

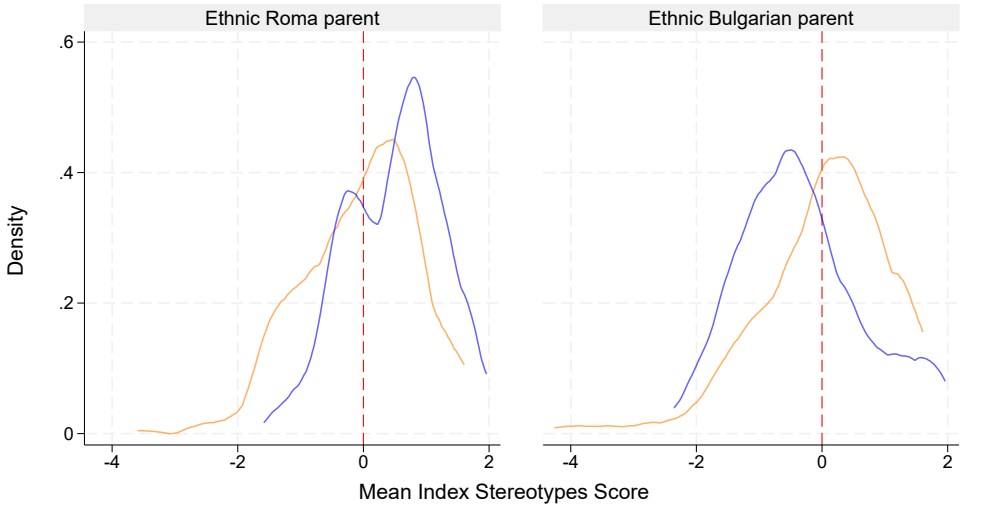


Figure 5: Distribution of parental stereotypes by ethnic group

nurturing environment at home to support them to do well at school; value education and school; they are hard-working at school; they are interested in learning; they are smart; they are disciplined; they are nice".

Table 17: Parental ethnic Bulgarian stereotypes and child's prosociality

	Child's prosociality (pooled)	Child's prosociality (pooled)	Child's prosociality (pooled)
Stereotypes towards ethnic Bulgarian	0.058	0.011	0.002
	(0.045)	(0.105)	(0.104)
Ethnic Bulgarian child		0.124	0.093
		(0.141)	(0.141)
Ethnic Bulgarian child × Stereotypes towards ethnic Bulgarian		0.059	0.067
		(0.114)	(0.113)
Parental prosociality (pooled)			0.108*
			(0.056)
Constant	0.456	0.461	0.345
	(0.622)	(0.620)	(0.608)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	842	842	842
Ethnic Bulgarian stereotypes and ethnic Bulgarian households - estimate		.0706	.0689
Ethnic Bulgarian stereotypes and ethnic Bulgarian households - p-value		.134	.134

The table presents the estimations of a linear regression of the average amount shared in the dictator game by the children on parents' stereotypes. The dependent variable indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each child during the game. Parental prosociality indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each parent during the game. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 18: Parental ethnic Bulgarian stereotypes and child's prosociality towards the out-group

	Child's prosociality	Child's prosociality	Child's prosociality
Stereotypes towards ethnic Bulgarian	0.061	-0.005	-0.011
	(0.048)	(0.111)	(0.111)
Ethnic Bulgarian child		0.117	0.078
		(0.140)	(0.144)
Ethnic Bulgarian child × Stereotypes towards ethnic Bulgarian		0.087	0.090
		(0.120)	(0.120)
Parental prosociality			0.115**
			(0.044)
Constant	-0.036	-0.027	-0.129
	(0.581)	(0.578)	(0.564)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	842	842	842
Ethnic Bulgarian stereotypes and ethnic Bulgarian households - estimate		.0814	.079
Ethnic Bulgarian stereotypes and ethnic Bulgarian households - p-value		.0922	.0975

The table presents the estimations of a linear regression of the amount shared in the dictator game by the children on parents' stereotypes. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the out-group receiver during the game. Parental prosociality indicates the average amount, standardized with respect to the whole sample, shared by each parent with the out-group receiver during the game. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 19: Parental ethnic Roma stereotypes and child's prosociality

	Child's prosociality (pooled)	Child's prosociality (pooled)	Child's prosociality (pooled)
Stereotypes towards ethnic Roma	0.089	-0.020	-0.043
	(0.054)	(0.140)	(0.138)
Ethnic Bulgarian child		0.083	0.047
		(0.159)	(0.161)
Ethnic Bulgarian child × Stereotypes towards ethnic Roma		0.146	0.151
		(0.144)	(0.141)
Parental prosociality (pooled)			0.101*
			(0.056)
Constant	0.374	0.518	0.442
	(0.641)	(0.641)	(0.628)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	842	842	842
Ethnic Roma stereotypes and ethnic Bulgarian households - estimate		.126	.108
Ethnic Roma stereotypes and ethnic Bulgarian households - p-value		.0137	.0373

The table presents the estimations of a linear regression of the average amount shared in the dictator game by the children on parents' stereotypes. The dependent variable indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each child during the game. Parental prosociality indicates the average amount, averaged between rounds and standardized with respect to the whole sample, shared by each parent during the game. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 20: Parental ethnic Roma stereotypes and child's prosociality towards the out-group

	Child's prosociality	Child's prosociality	Child's prosociality
Stereotypes towards ethnic Roma	0.166***	-0.002	-0.016
	(0.061)	(0.123)	(0.122)
Ethnic Bulgarian child		0.056	0.024
		(0.165)	(0.169)
Ethnic Bulgarian child × Stereotypes towards ethnic Roma		0.221	0.210
		(0.133)	(0.134)
Parental prosociality			0.096**
			(0.044)
Constant	-0.236	-0.008	-0.072
	(0.597)	(0.575)	(0.564)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	842	842	842
Ethnic Roma stereotypes and ethnic Bulgarian households - estimate		.219	.193
Ethnic Roma stereotypes and ethnic Bulgarian households - p-value		.000373	.00225

The table presents the estimations of a linear regression of the amount shared in the dictator game by the children on parents' stereotypes. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the out-group receiver during the game. Parental prosociality indicates the average amount, standardized with respect to the whole sample, shared by each parent with the out-group receiver during the game. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table 21: Parental stereotypes and intergenerational transmission of in-group preferences

	Children's in-group preference	Children's in-group preference
Parental in-group preference	0.035**	0.042**
	(0.017)	(0.018)
Ethnic Roma stereotypes	-0.355***	
	(0.119)	
Parental in-group preference × Ethnic Roma stereotypes	-0.001	
	(0.020)	
Ethnic Bulgarian stereotypes		-0.039
		(0.098)
Parental in-group preference × Ethnic Bulgarian stereotypes		-0.008
		(0.019)
Constant	0.821	0.726
	(1.064)	(1.051)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Observations	825	825
Intergen. assoc. of in-group preference among parents with more positive stereotypes - estimate	.0335	.034
Intergen. assoc. of in-group preference among parents with more positive stereotypes - p-value	.189	.229

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. The in-group preference variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among parents with more positive stereotypes towards ethnic Roma children across the whole sample. Standard errors clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

In ethnic Bulgarian households, stronger positive stereotypes towards ethnic Bulgarian children among parents are associated with an increase in children's prosociality towards the out-group receiver, as displayed in table 18. Similarly, the estimates in tables 19 and 20 show that, also in ethnic Bulgarian households, more positive parental stereotypes towards ethnic Roma children are associated with a relatively large and significant increase in child's prosociality on average (0.10 standard deviations, p-value < 0.05) and towards out-group receivers (0.19 standard deviations, p-value < 0.01). On the other hand, in ethnic Roma households, parental stereotypes are not associated with changes in children's prosocial behavior.

Moreover, I find a strong association between positive stereotypes towards Roma and the intergenerational association of prosociality in ethnic Bulgarian households. As shown in table D.7, the inclusion of the index of stereotypes towards ethnic Roma almost fully explains the transmission of prosociality towards the out-group found in table 9. Specifically, the intergenerational association coefficient on out-group prosociality becomes smaller and significant at the 10% level while the estimate for the Roma stereotypes index is large and statistically significant at the 1% level.

Finally, regarding the intergenerational association of in-group preferences – tables 21 and D.10 – I do not find any statistically significant association in the estimates across ethnic groups and measures of stereotypes. The only exception is a large and negative association of ethnic Roma stereotypes with children's in-group preferences in table 21. However, in that specification, the sample includes households from both ethnic groups. Therefore, it is difficult to assess the underlying process for this weakening of the intergenerational association of in-group preferences.

Stereotypes and out-group contact

Among ethnic Bulgarian parents, contact with the out group is not associated with changes in their stereotypes towards ethnic Bulgarian or ethnic Roma children. On the other hand, among ethnic Roma parents, contact with the out group is associated with more positive stereotypes towards ethnic Roma children (and marginally with more positive stereotypes towards ethnic Bulgarian children)[30]

These estimates are aligned with the negative inter-group contact hypothesis, a relatively recent development in social psychology (Paolini et al., 2010, 2024), which states that "negative contact makes individuals more aware of their respective group memberships" (Paolini et al., 2010). Therefore, negative contact may explain the association between inter-group contact by parents in Roma ethnic groups and stronger in-group stereotypes.

However, as in the previous section, I do not have information on the nature of inter-group interactions and I cannot test if these interactions had a negative, positive, or neutral valence. Therefore, those correlations should be understood as suggestive descriptive evidence.

5.2.4 Expectation of discrimination

Recent literature indicates that members of minority groups may embark on strategic behavior if they anticipate discrimination against them (Aksoy et al., 2023; Kudashvili & Lergetporer, 2022; Zussman, 2013). Therefore, in our setting, ethnic minority parents may anticipate that, in the future, their children will suffer discrimination due to their ethnicity and the prevalent discrimination against Roma individuals in Bulgaria (European Union Agency for Fundamental Rights, 2017). This expectation may

[30] Estimates shown in tables D.12 and D.13.

strengthen ethnic minority parents' social distance with respect to the ethnic majority, which may then be reflected in a lower prosociality towards the out-group.

To elicit their expectation of future discrimination against their child, each parent answered the following question "I believe that my child will be discriminated/treated differently as an adult when looking for a job because of his/her ethnicity if they stay in Bulgaria". All parents rated their subjective expectation on a scale between 0% and 100%, and were provided with a physical aid to help them state their expectation. The use of physical aids to elicit expectations and probabilities of future events is common in low- and middle-income contexts (Delavande et al., 2011).

Our data, shown in figure D.3, indicate that almost half of all Roma parents expect discrimination against their child with at least 50% probability. Conversely, only 24% of ethnic Bulgarian parents expect the same. Given the available evidence about the discrimination suffered by ethnic Roma individuals in Bulgaria, those estimates are not surprising. However, the interpretation of those estimates warrants certain caution: first, there is a relatively large share of non-responses in both ethnic groups: 21% of ethnic Roma and 12% of ethnic Bulgarian parents did not understand the question. Second, we have suggestive evidence that the instrument was not fully understood by several parents in our sample, as approximately only 60% pass the attention check. Therefore, given these data quality concerns, the corresponding estimates should be understood as suggestive evidence[31].

5.2.5 Parental and children's age and gender

Finally, I have also estimated the associations of parents' and children's age and gender in the intergenerational transmission of social preferences. All tables are displayed in B. Regarding the transmission of prosociality, I find a stronger intergenerational association of prosociality towards out-group receivers among older children in ethnic Bulgarian households and a stronger intergenerational association for older children's prosociality towards an in-group receiver in Roma households. I find no differences between younger and older parents in either ethnic Roma or ethnic Bulgarian households. Regarding the transmission of in-group preferences, I show evidence of intergenerational transmission of in-group preferences among Roma households with children of average age, and that children of older Roma parents have a weaker in-group preference than children of younger parents.

Female caregivers display different patterns of transmission between ethnic groups. In ethnic Roma households, female caregivers are associated with increases in the prosociality of their child towards the in-group receiver; in ethnic Bulgarian households, I find a weaker intergenerational association for female caregivers towards out-group receivers. This last result also emerges among children: weaker, yet positive and significant, intergenerational association for female children - compared to male children - towards out-group receivers in ethnic Bulgarian households.

Finally, I observe evidence of transmission of in-group preferences among both male and female children, weaker among female children, across the whole sample[32]. Among parents, I find intergenerational association of in-group preferences among female caregivers, but no evidence of transmission among male caregivers.

[31] See table D.11. Due to the described data concerns, I do not estimate the rest of the specifications.

[32] I fail to find evidence once I estimate the association separately by ethnic groups.

6 Discussion and conclusion

In this paper, I have studied the interaction between two important determinants of social preferences: (ethnic) group identity and parental social preferences. To do so, I have implemented a lab-in-the-field experiment in Bulgaria to collect the social preferences of 842 adults and children from two of the ethnic groups in the country, ethnic majority Bulgarians and ethnic minority Roma. In the experiment, each individual played two rounds of a standard dictator game, one round with a hypothetical ethnic in-group receiver and one round with a hypothetical ethnic out-group receiver. This design allows me to estimate (i) the association of social preferences with each player's ethnic identity at the individual level across ethnic groups and generations, and (ii) the differences in the intergenerational association of social preferences between ethnic groups.

My main findings provide novel evidence about potential determinants of social preferences during early childhood. I find that ethnic in-group bias in social preferences is associated between generations, a result not previously found in the literature, and that children also exhibit individual heterogeneity in their ethnic in-group bias, showing an empirical pattern previously found only among adults (Fong & Luttmer, 2009; Kranton et al., 2020). Moreover, I find suggestive evidence that the intergenerational association of social preferences, and its potentially explanatory mechanisms, may be different between ethnic groups.

However, an important open question remains: why are certain aspects of social preferences transmitted only in certain ethnic groups? I detect an intergenerational association of social preference in ethnic Bulgarian households, but not in ethnic Roma households. At the same time, I estimate that ethnic in-group bias is associated between generations on average, but I cannot reject a null association once I estimate the intergenerational association separately by ethnic group.

Regarding the former, I argue that this difference may emerge from how parents, in both ethnic groups, solve two different, but related, trade-offs regarding altruistic behavior and ethnic out-group contact. The first trade-off refers to their own behavior: Parents may value altruistic behavior, but acting altruistically could probably require interacting with an out-group member, which may generate disutility. The second trade-off refers to how they assess their child's behavior. Parents may also infer (dis)utility — following the idea of Beckerian altruism (Doepke & Zilibotti, 2017) — from how their children's solves the same trade-off. Then, depending on how each parent solves both trade-offs, different patterns of intergenerational transmission of prosociality may emerge. Unfortunately, I did not collect enough data to properly identify the described process. For example, I did not elicit how parents may assess their children's prosocial behavior towards individuals from the ethnic out-group. Therefore, future work should aim at collecting richer data to test the suggested mechanisms.

The null association of in-group preferences between generations within ethnic groups may be explained by a lack of statistical power[33]. First, when I estimate the in-group bias at the individual level, due to the experimental design, I observe one observation per individual instead of two. This reduction in the number of observations reduces the statistical power with respect to the rest of specifications. Moreover, the inclusion of community fixed effects restricts the existing differences between ethnic groups within communities that I can exploit. Out of the 115 communities in the sample, in only 37 of

[33]Note that a lack of statistical power may also play a role in the difference that I find between ethnic groups in the intergenerational association of social preferences. In those estimations, I leverage 665 observations in the ethnic Roma sample and 1,018 observations in the ethnic Bulgarian sample, and I cannot rule out the null hypothesis that these estimates are equal between both samples.

them I randomly sampled individuals from both ethnic Bulgarian and ethnic Roma groups. Therefore, the amount of actual variation between groups at the community level is limited, which may explain why I do not detect intergenerational association of in-group bias between ethnic groups.

Finally, another limitation of this study relates to the specific context where I collected data from. I elicit social preferences between ethnic groups in a context where the ethnic minority suffers prevalent instances of discrimination. These instances of discrimination could influence the emergence and display of social preferences in our setting. Hence, our estimates may not replicate in other settings where the relation between ethnic majority and minorities has a more positive valence. Future work should replicate this experimental design in other settings with larger samples to increase statistical power and to check if results replicate in contexts that have different inter-ethnic group relations.

References

Ahmed, A. M. (2010). What is in a surname? The role of ethnicity in economic decision making [Publisher: Routledge _eprint: https://doi.org/10.1080/00036840801964609]. *Applied Economics*, *42*(21), 2715–2723. https://doi.org/10.1080/00036840801964609

Aksoy, B., Chadd, I., & Koh, B. H. (2023). Sexual identity, gender, and anticipated discrimination in prosocial behavior. *European Economic Review*, *154*, 104427. https://doi.org/10.1016/j.euroecorev.2023.104427

Alesina, A., Miano, A., & Stantcheva, S. (2023). Immigration and Redistribution. *The Review of Economic Studies*, *90*(1), 1–39. https://doi.org/10.1093/restud/rdac011

Algan, Y., Beasley, E., Côté, S., Park, J., Tremblay, R. E., & Vitaro, F. (2022). The Impact of Childhood Social Skills and Self-Control Training on Economic and Noneconomic Outcomes: Evidence from a Randomized Experiment Using Administrative Data. *American Economic Review*, *112*(8), 2553–2579. https://doi.org/10.1257/aer.20200224

Algan, Y., & Cahuc, P. (2010). Inherited Trust and Growth. *American Economic Review*, *100*(5), 2060–2092. https://doi.org/10.1257/aer.100.5.2060

Algan, Y., & Cahuc, P. (2014, January). Chapter 2 - Trust, Growth, and Well-Being: New Evidence and Policy Implications. In P. Aghion & S. N. Durlauf (Eds.), *Handbook of Economic Growth* (pp. 49–120, Vol. 2). Elsevier. https://doi.org/10.1016/B978-0-444-53538-2.00002-2

Algan, Y., & Huillery, E. (2025). Socio-Emotional Skills and the Future of Education [Publisher: Annual Reviews]. *Annual Review of Economics*, *17*(Volume 17, 2025), 589–613. https://doi.org/10.1146/annurev-economics-081624-080702

Allport, G. W. (1954). *The nature of prejudice* [Pages: xviii, 537]. Addison-Wesley.

Altonji, J. G., & Blank, R. M. (1999, January). Chapter 48 Race and gender in the labor market. In *Handbook of Labor Economics* (pp. 3143–3259, Vol. 3). Elsevier. https://doi.org/10.1016/S1573-4463(99)30039-0

Angerer, S., Dutcher, E. G., Glätzle-Rützler, D., Lergetporer, P., & Sutter, M. (2017). Gender differences in discrimination emerge early in life: Evidence from primary school children in a bilingual city. *Economics Letters*, *152*, 15–18. https://doi.org/10.1016/j.econlet.2016.12.031

Angerer, S., Glätzle-Rützler, D., Lergetporer, P., & Sutter, M. (2016). Cooperation and discrimination within and across language borders: Evidence from children in a bilingual city. *European Economic Review*, *90*, 254–264. https://doi.org/10.1016/j.euroecorev.2016.02.022

Angerer, S., & Zhuravleva, T. (2024). Early childhood discrimination: A comparison of countries. *Journal of Behavioral and Experimental Economics*, *113*, 102301. https://doi.org/10.1016/j.socec.2024.102301

Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. https://doi.org/10.2307/j.ctvcm4j72

Balliet, D., Wu, J., & De Dreu, C. K. W. (2014). Ingroup favoritism in cooperation: A meta-analysis. *Psychological Bulletin*, *140*(6), 1556–1581. https://doi.org/10.1037/a0037737

Barron, K., Harmgart, H., Huck, S., Schneider, S. O., & Sutter, M. (2023). Discrimination, Narratives, and Family History: An Experiment with Jordanian Host and Syrian Refugee Children. *The Review of Economics and Statistics*, *105*(4), 1008–1016. https://doi.org/10.1162/rest_a_01090

Bartoš, V., & Levely, I. (2021). Sanctioning and trustworthiness across ethnic groups: Experimental evidence from Afghanistan. *Journal of Public Economics*, *194*, 104347. https://doi.org/10.1016/j.jpubeco.2020.104347

Bauer, M., Chytilová, J., & Pertold-Gebicka, B. (2014). Parental background and other-regarding preferences in children. *Experimental Economics*, *17*(1), 24–46. https://doi.org/10.1007/s10683-013-9355-y

Baumrind, D. (1967). Child care practices anteceding three patterns of preschool behavior [Place: US Publisher: Heldref Publications]. *Genetic Psychology Monographs*, *75*, 43–88.

Ben-Ner, A., List, J. A., Putterman, L., & Samek, A. (2017). Learned generosity? An artefactual field experiment with parents and their children. *Journal of Economic Behavior & Organization*, *143*, 28–44. https://doi.org/10.1016/j.jebo.2017.07.030

Berge, L. I. O., Bjorvatn, K., Galle, S., Miguel, E., Posner, D. N., Tungodden, B., & Zhang, K. (2020). Ethnically Biased? Experimental Evidence from Kenya. *Journal of the European Economic Association*, *18*(1), 134–164. https://doi.org/10.1093/jeea/jvz003

Bindra, P. C., Glätzle-Rützler, D., & Lergetporer, P. (2020). Discrimination at young age: Experimental evidence from preschool children. *Journal of Economic Behavior & Organization*, *175*, 55–70. https://doi.org/10.1016/j.jebo.2019.12.026

Bouckaert, J., & Dhaene, G. (2004). Inter-ethnic trust and reciprocity: Results of an experiment with small businessmen. *European Journal of Political Economy*, *20*(4), 869–886. https://doi.org/10.1016/j.ejpoleco.2003.08.006

Burns, J. (2012). Race, diversity and pro-social behavior in a segmented society. *Journal of Economic Behavior & Organization*, *81*(2), 366–378. https://doi.org/10.1016/j.jebo.2011.11.006

Bursztyn, L., & Yang, D. Y. (2022). Misperceptions About Others [_eprint: https://doi.org/10.1146/annurev-economics-051520-023322]. *Annual Review of Economics*, *14*(1), 425–452. https://doi.org/10.1146/annurev-economics-051520-023322

Cappelen, A., List, J., Samek, A., & Tungodden, B. (2020). The Effect of Early-Childhood Education on Social Preferences [Publisher: The University of Chicago Press]. *Journal of Political Economy*, *128*(7), 2739–2758. https://doi.org/10.1086/706858

Cetre, S., Algan, Y., Grimalda, G., Murtin, F., Pipke, D., Putterman, L., Schmidt, U., & Siegerink, V. (2024). Ethnic bias, economic achievement and trust between large ethnic groups: A study in Germany and the U.S. *Journal of Economic Behavior & Organization*, *224*, 996–1021. https://doi.org/10.1016/j.jebo.2024.07.003

Chen, Y., & Li, S. X. (2009). Group Identity and Social Preferences. *American Economic Review*, *99*(1), 431–457. https://doi.org/10.1257/aer.99.1.431

Chen, Y., Li, S. X., Liu, T. X., & Shih, M. (2014). Which hat to wear? Impact of natural identities on coordination and cooperation. *Games and Economic Behavior*, *84*, 58–86. https://doi.org/10.1016/j.geb.2013.12.002

Chetty, R., Jackson, M. O., Kuchler, T., Stroebel, J., Hendren, N., Fluegge, R. B., Gong, S., Gonzalez, F., Grondin, A., Jacob, M., Johnston, D., Koenen, M., Laguna-Muggenburg, E., Mudekereza, F., Rutter, T., Thor, N., Townsend, W., Zhang, R., Bailey, M., … Wernerfelt, N. (2022). Social capital I: Measurement and associations with economic mobility [Publisher: Nature Publishing Group]. *Nature*, *608*(7921), 108–121. https://doi.org/10.1038/s41586-022-04996-4

Chowdhury, S., Sutter, M., & Zimmermann, K. F. (2022). Economic Preferences across Generations and Family Clusters: A Large-Scale Experiment in a Developing Country [Publisher: The University of Chicago Press]. *Journal of Political Economy*, *130*(9), 2361–2410. https://doi.org/10.1086/720395

Cipriani, M., Giuliano, P., & Jeanne, O. (2013). Like mother like son? Experimental evidence on the transmission of values from parents to children. *Journal of Economic Behavior & Organization*, *90*, 100–111. https://doi.org/10.1016/j.jebo.2013.03.002

Cobb-Clark, D. A., Salamanca, N., & Zhu, A. (2019). Parenting style as an investment in human development [Publisher: Springer New York LLC]. *Journal of Population Economics*, *32*(4), 1315–1352. https://doi.org/10.1007/s00148-018-0703-2

Commission, E. (2020, October). Roma equality, inclusion and participation in the EU - European Commission. Retrieved August 7, 2026, from https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/combatting-discrimination/roma-eu/roma-equality-inclusion-and-participation-eu_en

Commission, E. (2024). Discrimination in the European Union - December 2023 - - Eurobarometer survey. Retrieved August 7, 2026, from https://europa.eu/eurobarometer/surveys/detail/2972

Commission, E. (n.d.). Bulgaria - European Commission. Retrieved August 7, 2026, from https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/combatting-discrimination/roma-eu/roma-equality-inclusion-and-participation-eu-country/bulgaria_en

Delavande, A., Giné, X., & McKenzie, D. (2011). Measuring subjective expectations in developing countries: A critical review and new evidence. *Journal of Development Economics*, *94*(2), 151–163. https://doi.org/10.1016/j.jdeveco.2010.01.008

Deming, D. J. (2017). The Growing Importance of Social Skills in the Labor Market*. *The Quarterly Journal of Economics*, *132*(4), 1593–1640. https://doi.org/10.1093/qje/qjx022

Doepke, M., & Zilibotti, F. (2017). Parenting With Style: Altruism and Paternalism in Intergenerational Preference Transmission [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA14634]. *Econometrica*, *85*(5), 1331–1371. https://doi.org/10.3982/ECTA14634

Dohmen, T., Falk, A., Huffman, D., & Sunde, U. (2012). The Intergenerational Transmission of Risk and Trust Attitudes. *The Review of Economic Studies*, *79*(2), 645–677. https://doi.org/10.1093/restud/rdr027

Durante, R., Mastrorocco, N., Minale, L., & Snyder, J. M. (2025). Unpacking Social Capital. *The Economic Journal*, *135*(667), 773–807. https://doi.org/10.1093/ej/ueae074

Elwert, F., Keller, T., & Kotsadam, A. (2023). Rearranging the Desk Chairs: A Large Randomized Field Experiment on the Effects of Close Contact on Interethnic Relations [Publisher: The

University of Chicago Press]. *American Journal of Sociology*, *128*(6), 1809–1840. https://doi.org/10.1086/724865

Engel, C. (2011). Dictator games: A meta study. *Experimental Economics*, *14*(4), 583–610. https://doi.org/10.1007/s10683-011-9283-7

Enke, B. (2024). Moral Boundaries [Publisher: Annual Reviews]. *Annual Review of Economics*, *16*(Volume 16, 2024), 133–157. https://doi.org/10.1146/annurev-economics-091223-093730

Ertac, S. (2020). The Formation and Malleability of Preferences and Noncognitive Skills. In K. F. Zimmermann (Ed.), *Handbook of Labor, Human Resources and Population Economics* (pp. 1–27). Springer International Publishing. https://doi.org/10.1007/978-3-319-57365-6_144-1

European Union Agency for Fundamental Rights (Ed.). (2017). *Second European Union minorities and discrimination survey: Main results*. Publications Office. https://doi.org/10.2811/902610

European Union Agency for Fundamental Rights (Ed.). (2025). *Rights of Roma and Travellers in 13 european countries: Perspectives from the Roma survey 2024*. Publications Office. https://doi.org/10.2811/5671307

Falk, A., Becker, A., Dohmen, T., Enke, B., Huffman, D., & Sunde, U. (2018). Global Evidence on Economic Preferences*. *The Quarterly Journal of Economics*, *133*(4), 1645–1692. https://doi.org/10.1093/qje/qjy013

Falk, A., Becker, A., Dohmen, T., Huffman, D., & Sunde, U. (2023). The Preference Survey Module: A Validated Instrument for Measuring Risk, Time, and Social Preferences [Publisher: INFORMS]. *Management Science*, *69*(4), 1935–1950. https://doi.org/10.1287/mnsc.2022.4455

Falk, A., Kosse, F., Pinger, P., Schildberg-Hörisch, H., & Deckers, T. (2021). Socioeconomic Status and Inequalities in Children's IQ and Economic Preferences [Publisher: The University of Chicago Press]. *Journal of Political Economy*, *129*(9), 2504–2545. https://doi.org/10.1086/714992

Fehr, E., Bernhard, H., & Rockenbach, B. (2008). Egalitarianism in young children [Number: 7208 Publisher: Nature Publishing Group]. *Nature*, *454*(7208), 1079–1083. https://doi.org/10.1038/nature07155

Fehr, E., & Charness, G. (2025). Social Preferences: Fundamental Characteristics and Economic Consequences. *Journal of Economic Literature*. Retrieved January 31, 2025, from https://www.aeaweb.org/articles?id=10.1257/jel.20241391&from=f

Fehr, E., Glätzle-Rützler, D., & Sutter, M. (2013). The development of egalitarianism, altruism, spite and parochialism in childhood and adolescence. *European Economic Review*, *64*, 369–383. https://doi.org/10.1016/j.euroecorev.2013.09.006

Ferraro, P. J., & Cummings, R. G. (2007). Cultural diversity, discrimination, and economic outcomes: An experimental analysis [Publisher: John Wiley & Sons, Ltd]. *Economic Inquiry*, *45*(2), 217–232. https://doi.org/10.1111/j.1465-7295.2006.00013.x

Fershtman, C., & Gneezy, U. (2001). Discrimination in a Segmented Society: An Experimental Approach. *The Quarterly Journal of Economics*, *116*(1), 351–377. https://doi.org/10.1162/003355301556338

Fong, C. M., & Luttmer, E. F. P. (2009). What Determines Giving to Hurricane Katrina Victims? Experimental Evidence on Racial Group Loyalty. *American Economic Journal: Applied Economics*, *1*(2), 64–87. https://doi.org/10.1257/app.1.2.64

Friesen, J., Arifovic, J., Wright, S. C., Ludwig, A., Giamo, L., & Baray, G. (2012). Ethnic identity and discrimination among children. *Journal of Economic Psychology*, *33*(6), 1156–1169. https://doi.org/10.1016/j.joep.2012.08.003

Giuliano, P., & Spilimbergo, A. (2025). Aggregate Shocks and the Formation of Preferences and Beliefs. *Journal of Economic Literature*, *63*(2), 542–597. https://doi.org/10.1257/jel.20241674

Haerpfer, C., Inglehart, R., Moreno, A., Welzel, C., Kizilova, K., Diez-Medrano, J., Lagos, M., Norris, P., Ponarin, E., & Puranen, B. (2022). World Values Survey Wave 7 (2017-2022) Cross-National Data-Set. https://doi.org/10.14281/18241.18

Haushofer, J., Lowes, S., Musau, A., Ndetei, D., Nunn, N., Poll, M., & Qian, N. (2023). Stress, Ethnicity, and Prosocial Behavior [Publisher: The University of Chicago Press]. *Journal of Political Economy Microeconomics*, *1*(2), 225–269. https://doi.org/10.1086/722367

Hett, F., Mechtel, M., & Kröll, M. (2020). The Structure and Behavioral Effects of Revealed Social Identity Preferences*. *The Economic Journal*, *130*(632), 2569–2595. https://doi.org/10.1093/ej/ueaa055

institute, N. .-. N. S. (2021). Census 2021 - Results. Retrieved August 7, 2026, from https://www.nsi.bg/en/statistical-data/151/1349

Kende, A., Hadarics, M., Bigazzi, S., Boza, M., Kunst, J. R., Lantos, N. A., Lášticová, B., Minescu, A., Pivetti, M., & Urbiola, A. (2021). The last acceptable prejudice in Europe? Anti-Gypsyism as the obstacle to Roma inclusion [Publisher: SAGE Publications Ltd]. *Group Processes & Intergroup Relations*, *24*(3), 388–410. https://doi.org/10.1177/1368430220907701

Kosse, F., Deckers, T., Pinger, P., Schildberg-Hörisch, H., & Falk, A. (2020). The Formation of Prosociality: Causal Evidence on the Role of Social Environment [Publisher: The University of Chicago Press]. *Journal of Political Economy*, *128*(2), 434–467. https://doi.org/10.1086/704386

Kosse, F., & Tincani, M. M. (2020). Prosociality predicts labor market success around the world [Publisher: Nature Publishing Group]. *Nature Communications*, *11*(1), 5298. https://doi.org/10.1038/s41467-020-19007-1

Kranton, R., Pease, M., Sanders, S., & Huettel, S. (2020). Deconstructing bias in social preferences reveals groupy and not-groupy behavior [Publisher: Proceedings of the National Academy of Sciences]. *Proceedings of the National Academy of Sciences*, *117*(35), 21185–21193. https://doi.org/10.1073/pnas.1918952117

Kudashvili, N., & Lergetporer, P. (2022). Minorities' strategic response to discrimination: Experimental evidence. *Journal of Public Economics*, *208*, 104630. https://doi.org/10.1016/j.jpubeco.2022.104630

Lane, T. (2016). Discrimination in the laboratory: A meta-analysis of economics experiments. *European Economic Review*, *90*, 375–402. https://doi.org/10.1016/j.euroecorev.2015.11.011

Levitt, S. D., & List, J. A. (2007). What Do Laboratory Experiments Measuring Social Preferences Reveal About the Real World? *Journal of Economic Perspectives*, *21*(2), 153–174. https://doi.org/10.1257/jep.21.2.153

Li, S. X. (2020). Group Identity, Ingroup Favoritism, and Discrimination. In *Handbook of Labor, Human Resources and Population Economics* (pp. 1–28). Springer, Cham. https://doi.org/10.1007/978-3-319-57365-6_123-1

List, A. J., List, J. A., & Samek, A. (2017). Discrimination among pre-school children: Field experimental evidence. *Economics Letters*, *157*, 159–162. https://doi.org/10.1016/j.econlet.2017.06.006

Lowe, M. (2025). Has Intergroup Contact Delivered? [Publisher: Annual Reviews]. *Annual Review of Economics*, *17*(Volume 17, 2025), 321–344. https://doi.org/10.1146/annurev-economics-081324-091109

Paolini, S., Gibbs, M., Sales, B., Anderson, D., & McIntyre, K. (2024). Negativity bias in intergroup contact: Meta-analytical evidence that bad is stronger than good, especially when people have the opportunity and motivation to opt out of contact [Place: US Publisher: American Psychological Association]. *Psychological Bulletin*, *150*(8), 921–964. https://doi.org/10.1037/bul0000439

Paolini, S., Harwood, J., & Rubin, M. (2010). Negative Intergroup Contact Makes Group Memberships Salient: Explaining Why Intergroup Conflict Endures [Publisher: SAGE Publications Inc]. *Personality and Social Psychology Bulletin*, *36*(12), 1723–1738. https://doi.org/10.1177/0146167210388667

Pettigrew, T. (1998). Intergroup Contact Theory. *Annual review of psychology*, *49*, 65–85. https://doi.org/10.1146/annurev.psych.49.1.65

Rao, G. (2019). Familiarity Does Not Breed Contempt: Generosity, Discrimination, and Diversity in Delhi Schools. *American Economic Review*, *109*(3), 774–809. https://doi.org/10.1257/aer.20180044

Ravetti, C., Sarr, M., Munene, D., & Swanson, T. (2019). Discrimination and favouritism among South African workers: Ethnic identity and union membership. *World Development*, *123*, 104604. https://doi.org/10.1016/j.worlddev.2019.06.027

Robayo-Abril, M., & Millan, N. (2019, March). *Breaking the Cycle of Roma Exclusion in the Western Balkans*. World Bank, Washington, DC. https://doi.org/10.1596/31393

Sutter, M., & Untertrifaller, A. (2020). Children's heterogeneity in cooperation and parental background: An experimental study. *Journal of Economic Behavior & Organization*, *171*, 286–296. https://doi.org/10.1016/j.jebo.2020.01.010

Tanaka, T., & Camerer, C. F. (2016). Trait perceptions influence economic out-group bias: Lab and field evidence from Vietnam. *Experimental Economics*, *19*(3), 513–534. https://doi.org/10.1007/s10683-015-9452-1

Van Der Merwe, W. G., & Burns, J. (2008). What's in a Name? Racial Identity and Altruism in Post-Apartheid South Africa [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1813-6982.2008.00174.x]. *South African Journal of Economics*, *76*(2), 266–275. https://doi.org/10.1111/j.1813-6982.2008.00174.x

Zhang, Z., Zhang, X., & Putterman, L. (2019). Trust and cooperation at a confluence of worlds: An experiment in Xinjiang, China. *Journal of Economic Behavior & Organization*, *161*, 128–144. https://doi.org/10.1016/j.jebo.2019.03.001

Zussman, A. (2013). Ethnic Discrimination: Lessons from the Israeli Online Market for Used Cars [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/ecoj.12059]. *The Economic Journal*, *123*(572), F433–F468. https://doi.org/10.1111/ecoj.12059

Appendix A Experimental procedures

Elicitation of social preferences - Parents

Now, in this section of the questionnaire, you will play a sharing game. There are no right or wrong choices in the game; whatever you decide to do it is fine.

Imagine that you receive 20 leva. We ask you to divide them between yourself and another Bulgarian majority parent from your town. You can assume that this parent has a similar level of material conditions as you. [Enumerator: present the showcard that represent the 20 leva]

You can divide them in any way you like: you can keep the 20 leva for yourself, split them in half, give the 20 leva to the other parent, etc.

1. How would you like to share the levas between yourself and the other parent?

[Enumerator: write down the number of levas allocated to the other parent]

Attention! be sure to read the game explanation

The second game has different conditions. It involves choosing one of two money distribution options, depending on how much money you and the other parent would receive.

Explain the game using any of the games on the showcard.

In one case (show option a) you receive bgn 5 and the other parent receives bgn 5. in the other case (show option b) you receive xx bgn and the other parent receives xx bgn.

Show card with the 4 games in this game

I remind you that you are playing with a parent of Bulgarian ethnic origin

2. Please, for each pair of allocations, choose which one you prefer the most for each row.

[Enumerator: present the showcard that represent the different allocations]

	Allocation A	**Allocation B**	**Please, circle which allocation do you prefer**
1	You receive 5 leva The other parent receive 5 leva	You receive 10 leva The other parent receive 0 leva	A B
2	You receive 5 leva The other parent receive 5 leva	You receive 5 leva The other parent receive 0 leva	A B
3	You receive 5 leva The other parent receive 5 leva	You receive 10 leva The other parent receive 15 leva	A B
4	You receive 5 leva The other parent receive 5 leva	You receive 5 leva The other parent receive 10 leva	A B

Now, imagine that you receive 20 leva again. This time, we ask you to divide them between yourself and another Roma parent from your town. You can assume that this parent has a similar level of material conditions as you.

You can divide them in any way you like: you can keep the 20 leva for yourself, split them in half, give the 20 leva to the other parent, etc.

3. How would you like to share the levas between yourself and the other parent?

[Enumerator: write down the number of levas allocated to the other parent]

Attention! be sure to read the game explanation

The second game has different conditions. It involves choosing one of two money distribution options, depending on how much money you and the other parent would receive.

Explain the game using any of the games on the showcard.

In one case (show option a) you receive bgn 5 and the other parent receives bgn 5. in the other case (show option b) you receive xx bgn and the other parent receives xx bgn.

Show card with the 4 games in this game

I remind you that you are playing with a parent of Roma ethnic origin

4. Please, for each pair of allocations, choose which one you prefer the most for each row.

	Allocation A	**Allocation B**	**Please, circle which allocation do you prefer**
1	You receive 5 leva The other parent receive 5 leva	You receive 10 leva The other parent receive 0 leva	A B
2	You receive 5 leva The other parent receive 5 leva	You receive 5 leva The other parent receive 0 leva	A B
3	You receive 5 leva The other parent receive 5 leva	You receive 10 leva The other parent receive 15 leva	A B
4	You receive 5 leva The other parent receive 5 leva	You receive 5 leva The other parent receive 10 leva	A B

Elicitation of social preferences - Children

Now, we will play some games. In those games, there are no wrong or right answers. Whatever you decide to do is fine.

[Enumerator: place the 10 stickers and the drawing that represents the Bulgarian majority child in front of the child] *[Pointing at the drawing]* Imagine that you will be playing the game with this child. Imagine that this child is a child like you. His name is Nikola and comes from a non-Roma family.

The game is a sharing game. You will play with 10 stickers that you will need to share between you and Nikola. You can divide them in any way you like: you can keep the 10 stickers for yourself, split them in half, give the 10 stickers to Nikola, etc. As I said, there are no right or wrong answers. Whatever you decide to do is fine.

1. How many stickers would you like to share with Nikola? Circle them in the showcard.

[Enumerator: record on the tablet the number of stickers circled in the showcard] *[Once the child has circled the stickers, thank the kid (but do not prompt or praise him for their choice) and collect the showcard]* We will continue the game with Nikola. Now, we have defined a few extra pairs of allocations of stickers between you and him.

[Place the showcard indicated by the tablet in front of the child] We have defined a white split and a gray split of stickers. As you can see, on the white split, you receive XX stickers and Nikola receives YY stickers, and in the gray split you receive ZZ stickers and Nikola receives TT stickers. You can decide which of the two splits - white or gray - you prefer. There is no right or wrong answer. Whatever you decide, it is fine.

[Check for understanding] If you pick the white split, how many stickers would Nikola receive? [wait for the child to answer] And you?

[If the child does not answer correctly, repeat the explanation] *[If the child answers correctly, proceed with the game]*

2. Now, which of these two splits you prefer?

	Allocation A	**Allocation B**	**Please, circle which allocation do you prefer**
1	You receive 5 sticker The other child receives 5 stickers	You receive 10 stickers The other child receives 0 stickers	A B
2	You receive 5 stickers The other child receives 5 stickers	You receive 5 stickers The other child receives 0 stickers	A B
3	You receive 5 stickers The other child receives 5 stickers	You receive 10 stickers The other child receives 15 stickers	A B
4	You receive 5 stickers The other child receives 5 stickers	You receive 5 stickers The other child receives 10 stickers	A B

[Enumerator: record on the tablet the preferred allocation and show the next showcard indicated by the tablet. Repeat the process above for the remaining showcards]

[Once a child has made a decision on the four different splits, collect the showcards and the image of Nikola, and place the image of Martin in front of the child]

Now, we will play the same two games again. But this time imagine that you are playing with this child. Imagine that this child is a child like you. His name is Martin and comes from a Roma family.

You have 10 stickers again. [Place the stickers in front of the child] and we ask you to share them with Martin. Remember that you can divide them in any way you like: you can keep the 10 stickers for yourself, split them in half, give the 10 stickers to the other child, etc. and that any choice you make is fine.

3. How many stickers would you like to share with Martin? Circle them in the showcard.

[Enumerator: record on the tablet the number of stickers circled in the showcard]

[Once the child has circled the stickers, thank the kid (but do not prompt or praise him for their choice) and collect the showcard]

We will continue the game with Martin. As before, we have defined a few extra pairs of allocations of stickers between you and him.

[Place the showcard indicated by the tablet in front of the child]

We have defined a white split and a gray split of stickers. As you can see, on the white split, you receive XX stickers and Martin receives YY stickers, and in the gray split you receive ZZ stickers and Martin receives TT stickers.

You can decide which of the two splits - white or gray - you prefer. There is no right or wrong answer. Whatever you decide, it is fine.

4. Now, which of these two splits you prefer?

[Enumerator: record on the tablet the preferred allocation and show the next showcard indicated by the tablet. Repeat the process above for the remaining showcards] [Once a child has made a decision on the four different splits, collect the showcards and the image of Martin, and thank the kid for their time and for completing the survey]

	Allocation A	**Allocation B**	**Please, circle which allocation do you prefer**
1	You receive 5 sticker The other child receives 5 stickers	You receive 10 stickers The other child receives 0 stickers	A B
2	You receive 5 stickers The other child receives 5 stickers	You receive 5 stickers The other child receives 0 stickers	A B
3	You receive 5 stickers The other child receives 5 stickers	You receive 10 stickers The other child receives 15 stickers	A B
4	You receive 5 stickers The other child receives 5 stickers	You receive 5 stickers The other child receives 10 stickers	A B

Figure A.1: Children's allocation

Figure A.2: Parents' allocation

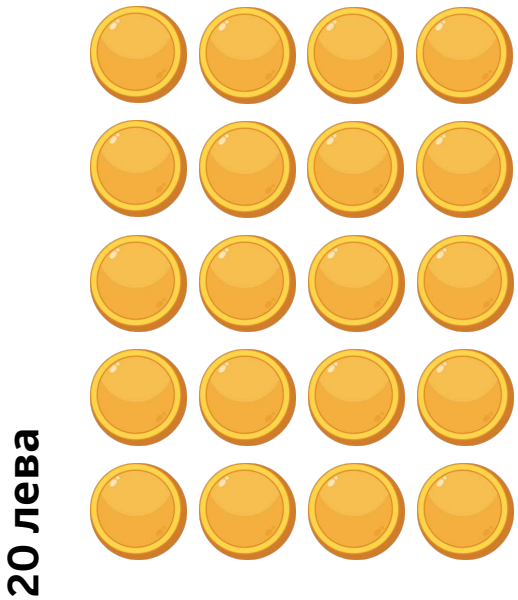


Figure A.3: Modified dictator game - showcard allocation



Figure A.4: Nikola (ethnic Bulgarian)



Никола

Figure A.5: Martin (ethnic Roma)



Мартин

Appendix B Parental and children's age and gender as mechanisms of intergenerational transmission of social preferences

Table B.1: Parents' age and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality
In-group receiver	0.074	0.195***
	(0.064)	(0.047)
Parental prosociality	0.108	0.145***
	(0.104)	(0.050)
In-group receiver × Parental prosociality	-0.001	-0.111*
	(0.069)	(0.058)
Parent's age	0.042	0.035
	(0.065)	(0.044)
In-group receiver × Parent's age	-0.130**	-0.043
	(0.055)	(0.041)
Parental prosociality × Parent's age	0.039	0.000
	(0.081)	(0.043)
In-group receiver × Parental prosociality × Parent's age	-0.064	0.008
	(0.066)	(0.069)
Constant	0.571***	0.314***
	(0.115)	(0.060)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018
Parental age and in-group receiver - estimate	-.0874	-.00733
Parental age and in-group receiver - p-value	.197	.877
Intergenerational association for parental age - estimate	.147	.145
Intergenerational association for parental age - p-value	.388	.0136
Intergenerational association for parental age and in-group receiver - estimate	.082	.0424
Intergenerational association for parental age and in-group receiver - p-value	.608	.53
Equality of coefficients	.924	.924

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. First column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver corresponds to the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. Age is a standardized variable that indicates parental age. First interaction assesses if parental age is associated with child's prosociality when interacting with in-group receivers. Second interaction assesses the intergenerational association estimate for older parents with the out-group. Third interaction assesses the intergenerational association estimate towards in-group receivers for older parents. Equality of coefficients presents the p-value of the null hypothesis that the association between parental age and child's prosociality is the same between ethnic groups. Standard errors clustered at the community level. We include community fixed effects. We control for age and gender of each parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.2: Children's age and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality
In-group receiver	0.109*	0.186***
	(0.060)	(0.049)
Parental prosociality	0.088	0.143***
	(0.084)	(0.049)
In-group receiver × Parental prosociality	-0.001	-0.104**
	(0.064)	(0.049)
Child's age	0.072	0.020
	(0.065)	(0.057)
In-group receiver × Child's age	0.025	-0.030
	(0.057)	(0.050)
Parental prosociality × Child's age	0.024	0.035
	(0.051)	(0.047)
In-group receiver × Parental prosociality × Child's age	0.109**	0.010
	(0.051)	(0.053)
Constant	0.452***	0.315***
	(0.143)	(0.061)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018
Child's age and in-group receiver - estimate	.0968	-.00974
Child's age and in-group receiver - p-value	.234	.847
Intergenerational association for child's age - estimate	.112	.177
Intergenerational association for child's age - p-value	.195	.0159
Intergenerational association for child's age and in-group receiver - estimate	.219	.0833
Intergenerational association for child's age and in-group receiver - p-value	.0702	.368
Equality of coefficients	.505	.505

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The first column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver corresponds to the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. Age is a standardized variable that indicates the child's age. The first interaction assesses if child's age is associated with child's prosociality when interacting with in-group receivers. The second interaction assesses the intergenerational association estimate for older children with the out-group. The third interaction assesses the intergenerational association estimate towards in-group receivers for older children. Equality of coefficients presents the p-value of the null hypothesis that the association between child's age and child's prosociality is the same between ethnic groups. Standard errors are clustered at the community level. We include community fixed effects. We control for age and gender of each parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.3: Parents' gender and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality
In-group receiver	0.032	0.124
	(0.071)	(0.079)
Parental prosociality	0.047	0.163**
	(0.089)	(0.064)
In-group receiver × Parental prosociality	0.090	-0.068
	(0.081)	(0.099)
Female	0.222	0.105
	(0.227)	(0.164)
In-group receiver × Female	0.138*	0.096
	(0.080)	(0.083)
Female × Parental prosociality	0.086	-0.030
	(0.092)	(0.077)
In-group receiver × Female × Parental prosociality	-0.138	-0.060
	(0.150)	(0.135)
Constant	0.591***	0.344***
	(0.124)	(0.063)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018
Female caregivers and in-group receiver - estimate	.36	.202
Female caregivers and in-group receiver - p-value	.0677	.197
Intergenerational association for female caregivers - estimate	.133	.133
Intergenerational association for female caregivers - p-value	.203	.0229
Intergenerational association for female caregivers and in-group receiver - estimate	.0841	.00565
Intergenerational association for female caregivers and in-group receiver - p-value	.418	.948
Equality of coefficients	.648	.648

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The first column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver corresponds to the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. Female is a standardized variable that indicates female caregivers. The first interaction assesses if caregiver's gender is associated with child's prosociality when interacting with in-group receivers. The second interaction assesses the intergenerational association estimate for female caregivers with the out-group. The third interaction assesses the intergenerational association estimate towards in-group receivers for female caregivers. Equality of coefficients presents the p-value of the null hypothesis that the association between female caregivers and child's prosociality is the same between ethnic groups. Standard errors are clustered at the community level. We include community fixed effects. We control for age and gender of each parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.4: Children's gender and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality
In-group receiver	0.096	0.174**
	(0.070)	(0.085)
Parental prosociality	0.048	0.187***
	(0.090)	(0.057)
In-group receiver × Parental prosociality	0.082	-0.123
	(0.063)	(0.078)
Female	-0.197	-0.221
	(0.200)	(0.149)
In-group receiver × Female	0.021	0.018
	(0.075)	(0.113)
Female × Parental prosociality	0.098	-0.090
	(0.093)	(0.061)
In-group receiver × Female × Parental prosociality	-0.148	0.041
	(0.156)	(0.125)
Constant	0.563***	0.325***
	(0.125)	(0.065)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018
Female children and in-group receiver - estimate	-.176	-.203
Female children and in-group receiver - p-value	.348	.126
Intergenerational association for female children - estimate	.146	.0971
Intergenerational association for female children - p-value	.159	.0695
Intergenerational association for female children and in-group receiver - estimate	.081	.0144
Intergenerational association for female children and in-group receiver - p-value	.515	.882
Equality of coefficients	.914	.914

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The first column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver corresponds to the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. Female is a standardized variable that indicates female children. The first interaction assesses if child's gender is associated with child's prosociality when interacting with in-group receivers. The second interaction assesses the intergenerational association estimate for female children with the out-group. The third interaction assesses the intergenerational association estimate towards in-group receivers for female children. Equality of coefficients presents the p-value of the null hypothesis that the association between female children and child's prosociality is the same between ethnic groups. Standard errors are clustered at the community level. We include community fixed effects. We control for age and gender of each parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.5: Parents' age and intergenerational transmission of in-group preference

	Children's in-group preference	Children's in-group preference	Children's in-group preference	Children's in-group preference
Parental in-group preference	0.042**	0.043**	0.066	0.068*
	(0.018)	(0.018)	(0.040)	(0.041)
Parent's age	-0.106	-0.106	-0.251**	-0.250**
	(0.075)	(0.076)	(0.116)	(0.117)
Parental in-group preference × Parent's age	-0.007	-0.006	0.011	0.013
	(0.012)	(0.012)	(0.038)	(0.038)
Ethnic Bulgarian household			0.005	0.012
			(0.311)	(0.317)
Ethnic Bulgarian household × Parental in-group preference			-0.033	-0.034
			(0.050)	(0.051)
Ethnic Bulgarian household × Parent's age			0.235*	0.233*
			(0.135)	(0.136)
Ethnic Bulgarian household × Parental in-group preference × Parent's age			-0.016	-0.018
			(0.040)	(0.041)
Constant	0.331***	0.236***	0.293	0.201
	(0.009)	(0.083)	(0.190)	(0.205)
Children's demographic variables	No	Yes	No	Yes
Parents' demographic variables	No	Yes	No	Yes
Observations	825	825	825	825
Intergen. assoc. of in-group preference among older parents - estimate	.0348	.0363		
Intergen. assoc. of in-group preference among older parents - p-value	.048	.0471		
Intergen. assoc. of in-group preference among older Roma parents - estimate			.0767	.0811
Intergen. assoc. of in-group preference among older Roma parents - p-value			.233	.215
Intergen. assoc. of in-group preference among older ethnic Bulgarian parents - estimate			.0279	.0292
Intergen. assoc. of in-group preference among older ethnic Bulgarian parents - p-value			.187	.182

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. Age is a standardized variable of parents' age. The in-group preference variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among older parents across the whole sample and the different ethnic groups. Standard errors are clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.6: Children's age and intergenerational transmission of in-group preference

	Children's in-group preference	Children's in-group preference	Children's in-group preference	Children's in-group preference
Parental in-group preference	0.041**	0.043**	0.072**	0.072**
	(0.018)	(0.018)	(0.035)	(0.035)
Child's age	0.009	0.006	0.065	0.055
	(0.092)	(0.092)	(0.139)	(0.138)
Parental in-group preference × Child's age	-0.017	-0.017	-0.035	-0.036
	(0.017)	(0.017)	(0.029)	(0.028)
Ethnic Bulgarian household			-0.024	-0.006
			(0.305)	(0.313)
Ethnic Bulgarian household × Parental in-group preference			-0.040	-0.037
			(0.044)	(0.044)
Ethnic Bulgarian household × Child's age			-0.078	-0.067
			(0.144)	(0.144)
Ethnic Bulgarian household × Parental in-group preference × Child's age			0.021	0.023
			(0.034)	(0.034)
Constant	0.328***	0.229***	0.328*	0.225
	(0.010)	(0.083)	(0.185)	(0.201)
Children's demographic variables	No	Yes	No	Yes
Parents' demographic variables	No	Yes	No	Yes
Observations	825	825	825	825
Intergen. assoc. of in-group preference among older children - estimate	.0234	.0261		
Intergen. assoc. of in-group preference among older children - p-value	.288	.241		
Intergen. assoc. of in-group preference among Roma parents with older children - estimate			.0366	.036
Intergen. assoc. of in-group preference among Roma parents with older children - p-value			.452	.454
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with older children - estimate			.0174	.022
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with older children - p-value			.507	.418

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. Age is a standardized variable of the child's age. The in-group preference variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among older children across the whole sample and the different ethnic groups. Standard errors are clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.7: Parents' gender and intergenerational transmission of in-group preference

	Children's in-group preference	Children's in-group preference	Children's in-group preference	Children's in-group preference
Parental in-group preference	0.045	0.048*	0.066	0.071
	(0.028)	(0.028)	(0.048)	(0.048)
Female	0.162	0.320	0.287	0.454
	(0.134)	(0.308)	(0.198)	(0.348)
Female × Parental in-group preference	-0.007	-0.009	-0.011	-0.016
	(0.026)	(0.026)	(0.049)	(0.050)
Ethnic Bulgarian household			0.109	0.124
			(0.321)	(0.324)
Ethnic Bulgarian household × Parental in-group preference			-0.030	-0.033
			(0.063)	(0.064)
Ethnic Bulgarian household × Female			-0.226	-0.221
			(0.276)	(0.276)
Ethnic Bulgarian household × Female × Parental in-group preference			0.005	0.009
			(0.061)	(0.062)
Constant	0.230***	0.229***	0.167	0.156
	(0.084)	(0.084)	(0.193)	(0.198)
Children's demographic variables	No	Yes	No	Yes
Parents' demographic variables	No	Yes	No	Yes
Observations	825	825	825	825
Intergen. assoc. of in-group preference among female parents - estimate	.0383	.0391		
Intergen. assoc. of in-group preference among female parents - p-value	.0372	.0317		
Intergen. assoc. of in-group preference among female Roma parents - estimate			.0554	.0554
Intergen. assoc. of in-group preference among female Roma parents - p-value			.2	.203
Intergen. assoc. of in-group preference among female ethnic Bulgarian parents - estimate			.0305	.0317
Intergen. assoc. of in-group preference among female ethnic Bulgarian parents - p-value			.182	.164

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. Female is a standardized variable of the parent's gender. The in-group preference variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among female parents across the whole sample and the different ethnic groups. Standard errors are clustered at the community level. We add community fixed effects. We control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table B.8: Children's gender and intergenerational transmission of in-group preference

	Children's in-group preference	Children's in-group preference	Children's in-group preference	Children's in-group preference
Parental in-group preference	0.048*	0.051*	0.079	0.084
	(0.029)	(0.028)	(0.059)	(0.058)
Female	0.049	-0.182	0.154	-0.091
	(0.155)	(0.325)	(0.176)	(0.289)
Female × Parental in-group preference	-0.014	-0.015	-0.034	-0.040
	(0.029)	(0.027)	(0.065)	(0.064)
Ethnic Bulgarian household			0.049	0.052
			(0.323)	(0.327)
Ethnic Bulgarian household × Parental in-group preference			-0.042	-0.045
			(0.071)	(0.070)
Ethnic Bulgarian household × Female			-0.181	-0.146
			(0.303)	(0.296)
Ethnic Bulgarian household × Female × Parental in-group preference			0.025	0.033
			(0.075)	(0.074)
Constant	0.304***	0.228***	0.272	0.194
	(0.079)	(0.084)	(0.182)	(0.202)
Children's demographic variables	No	Yes	No	Yes
Parents' demographic variables	No	Yes	No	Yes
Observations	825	825	825	825
Intergen. assoc. of in-group preference among female children - estimate	.0339	.0359		
Intergen. assoc. of in-group preference among female children - p-value	.0474	.0385		
Intergen. assoc. of in-group preference among Roma parents with female children - estimate			.0454	.0441
Intergen. assoc. of in-group preference among Roma parents with female children - p-value			.277	.285
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with female children - estimate			.0282	.0317
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with female children - p-value			.178	.147

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. Female is a standardized variable of the child's gender. The in-group preference variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among female children across the whole sample and the different ethnic groups. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Appendix C Main specifications without community fixed effects

Table C.1: Ethnic identity influence on parental social preferences - without community fixed effects

	Parental prosociality	Parental prosociality	Parental prosociality	Parental prosociality
In-group receiver	0.112***	0.112***	0.237***	0.237***
	(0.039)	(0.039)	(0.043)	(0.043)
Ethnic Bulgarian sender		0.122	0.225*	0.262**
		(0.102)	(0.114)	(0.116)
Ethnic Bulgarian sender × In-group receiver			-0.207***	-0.207***
			(0.069)	(0.069)
Constant	-0.056	-0.130	-0.192**	0.046
	(0.055)	(0.093)	(0.095)	(0.144)
Parents' demographic variables	No	No	No	Yes
Observations	1683	1683	1683	1683
Receiver's in-group interaction - point estimate			.0304	.0304
Receiver's in-group interaction - p-value			.569	.569

The table presents the linear regression estimates of parental prosociality on receiver and sender's ethnicity. The dependent variable indicates the amount, standardized across individuals and rounds, shared by each parent with the receiver during each round of the game. Ethnic Bulgarian sender is a bivariate variable that equals 1 if the sender identifies as ethnic Bulgarian and equals 0 if the sender identifies as ethnic Roma. In-group receiver is a bivariate variable that equals 1 if the game's receiver is from the same ethnic group as the sender, and equals 0 otherwise. Receiver's in-group interaction tests the null hypothesis that the partial association of increased parental prosociality when interacting with an in-group receiver equals zero for ethnic Bulgarian senders. Standard errors are clustered at the community level. I control for age and gender of each player. I do not include community fixed effects. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table C.2: Ethnic identity influence on children's social preferences - without community fixed effects

	Child's prosociality	Child's prosociality	Child's prosociality	Child's prosociality
In-group receiver	0.168***	0.168***	0.144**	0.144**
	(0.038)	(0.038)	(0.060)	(0.060)
Ethnic Bulgarian sender		0.117	0.098	0.085
		(0.090)	(0.096)	(0.101)
Ethnic Bulgarian sender × In-group receiver			0.039	0.039
			(0.075)	(0.075)
Constant	-0.084*	-0.155*	-0.143	-0.245*
	(0.048)	(0.090)	(0.092)	(0.128)
Children's demographic variables	No	No	No	Yes
Observations	1684	1684	1684	1684
Receiver's in-group interaction - point estimate			.183	.183
Receiver's in-group interaction - p-value			.000152	.000154

The table presents the linear regression estimates of children's prosociality on receiver and sender's ethnicity. The dependent variable indicates the amount, standardized across individuals and rounds, shared by each child with the receiver during each round of the game. Ethnic Bulgarian sender is a bivariate variable that equals 1 if the sender identifies as ethnic Bulgarian and equals 0 if the sender identifies as ethnic Roma. In-group receiver is a bivariate variable that equals 1 if the game's receiver is from the same ethnic group as the sender, and equals 0 otherwise. Receiver's in-group interaction tests the null hypothesis that the partial association of increased children's prosociality when interacting with an in-group receiver equals zero for ethnic Bulgarian senders. Standard errors are clustered at the community level. I control for age and gender of each player. I do not include community fixed effects. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table C.3: Intergenerational transmission of social preferences - without community fixed effects

	Child's prosociality (pooled)	Child's prosociality (pooled)
Parental prosociality (pooled)	0.163***	0.156***
	(0.052)	(0.053)
Constant	0.000	-1.291**
	(0.047)	(0.494)
Children's demographic variables	No	Yes
Parents' demographic variables	No	Yes
Observations	842	842

The table presents linear regression estimates of the intergenerational association between parents' and children's prosociality. We pool the amount shared by each child and their parents across their respective rounds of the dictator game and then standardize with respect to their samples. The dependent variable indicates the average amount across rounds shared by each child during the dictator game, and the independent variable is the average amount across rounds shared by their parent. Both variables are standardized with respect to their samples. Standard errors are clustered at the community level. I do not include community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table C.4: Intergenerational transmission of prosociality within ethnic groups - without community fixed effects

	Child's prosociality	Child's prosociality
In-group receiver	0.100*	0.183***
	(0.058)	(0.047)
Parental prosociality	0.142	0.158***
	(0.085)	(0.045)
In-group receiver × Parental prosociality	0.029	-0.081*
	(0.069)	(0.041)
Constant	-2.116***	-0.578
	(0.658)	(0.422)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1,018
Joint interaction test - estimate	.171	.077
Joint interaction test - p-value	.1	.21
Equality of coefficients	.87	.87

The table presents the linear regression estimates of child's prosociality on parental prosociality and receiver's ethnicity. The first column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver is from the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. The joint interaction tests, separately for each ethnic group, the null hypothesis of null intergenerational association when interacting with an in-group receiver. The equality of coefficients row reports the p-value for the null hypothesis of an equal intergenerational association estimate between ethnic groups. Standard errors are clustered at the community level. I do not include community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table C.5: Intergenerational transmission of in-group preferences - without community fixed effects

	Child's in-group preference	Child's in-group preference	Child's in-group preference	Child's in-group preference
Parental in-group preference	0.038	0.041*	0.082**	0.081**
	(0.025)	(0.025)	(0.040)	(0.039)
Ethnic Bulgarian household			0.191	0.252
			(0.158)	(0.162)
Ethnic Bulgarian household × Parental in-group preference			-0.055	-0.048
			(0.052)	(0.051)
Constant	0.340***	0.670	0.206	0.543
	(0.078)	(1.013)	(0.125)	(1.020)
Children's demographic variables	No	Yes	No	Yes
Parents' demographic variables	No	Yes	No	Yes
Observations	841	841	841	841
Sender's ethnicity interaction - point estimate			.0269	.0331
Sender's ethnicity interaction - p-value			.393	.293

The table presents the linear regression estimates of the intergenerational association between parents' and children's in-group preferences. The dependent variable is the child's in-group preference - the difference between the amount allocated to the in-group receiver and the out-group receiver during the dictator game - and the main independent variable is the parental in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. The variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The joint interaction tests the null hypothesis of a null intergenerational association of in-group preferences in ethnic Bulgarian households. Standard errors are clustered at the community level. I do not include community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Appendix D Additional tables and figures

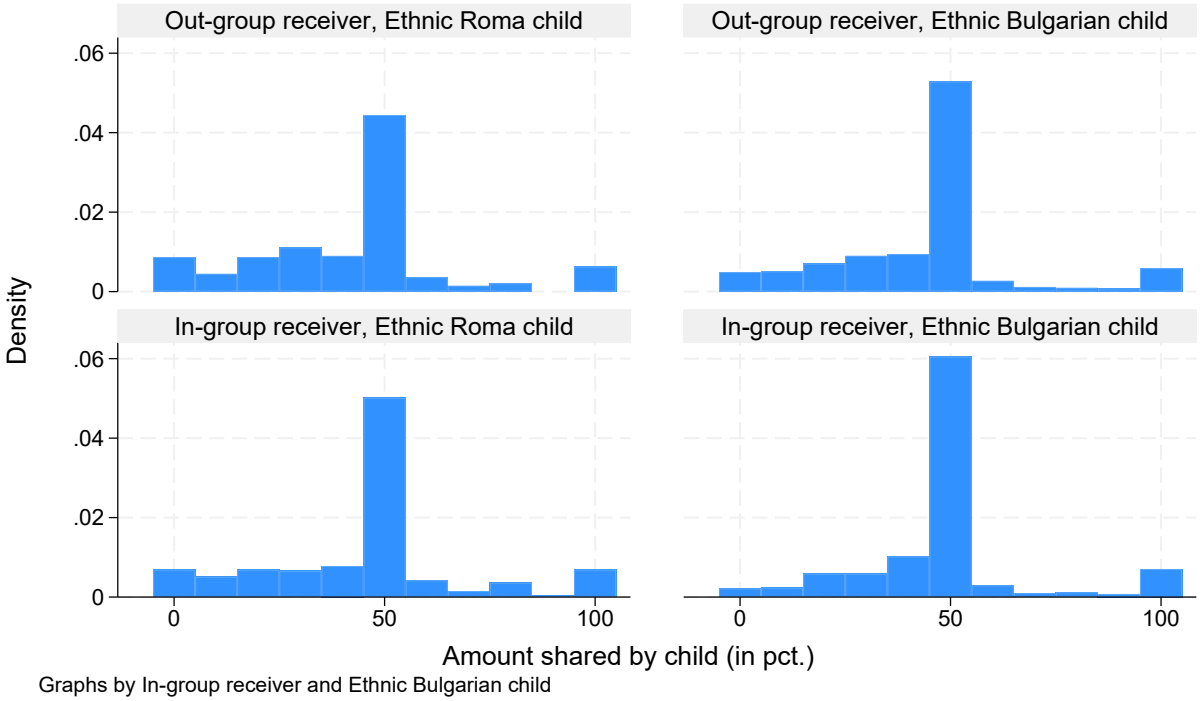


Figure D.1: Distribution of children's giving by ethnic group and receiver

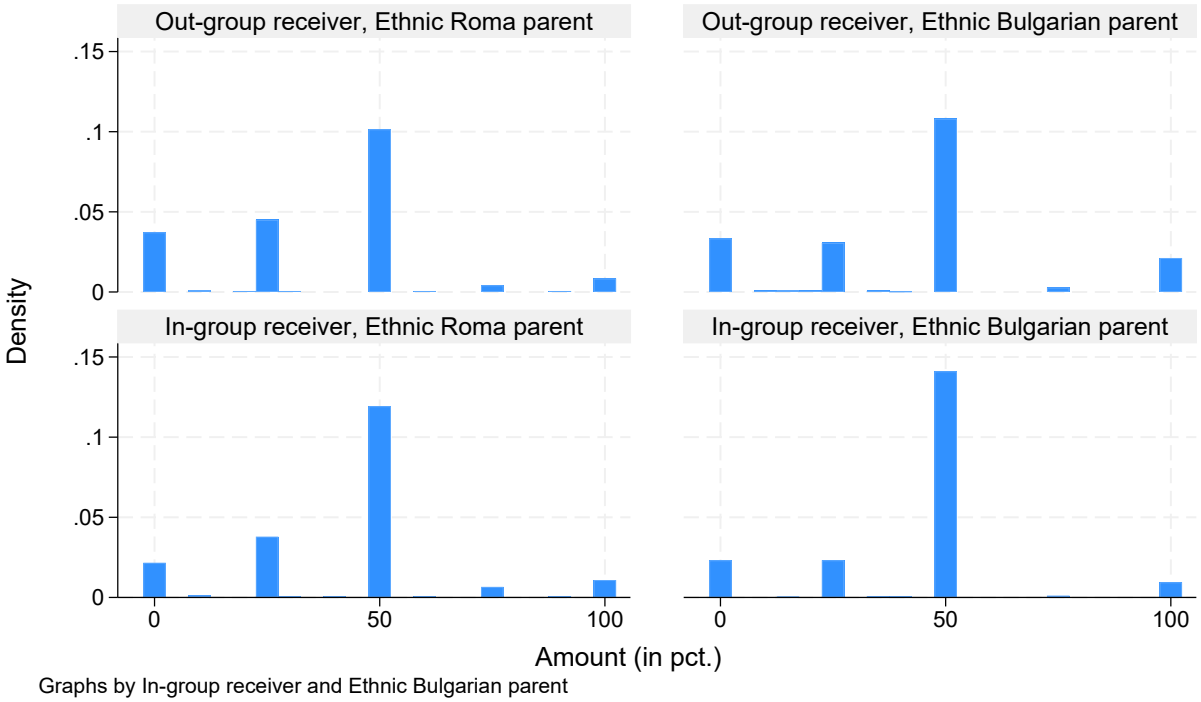


Figure D.2: Distribution of parental giving by ethnic group and receiver

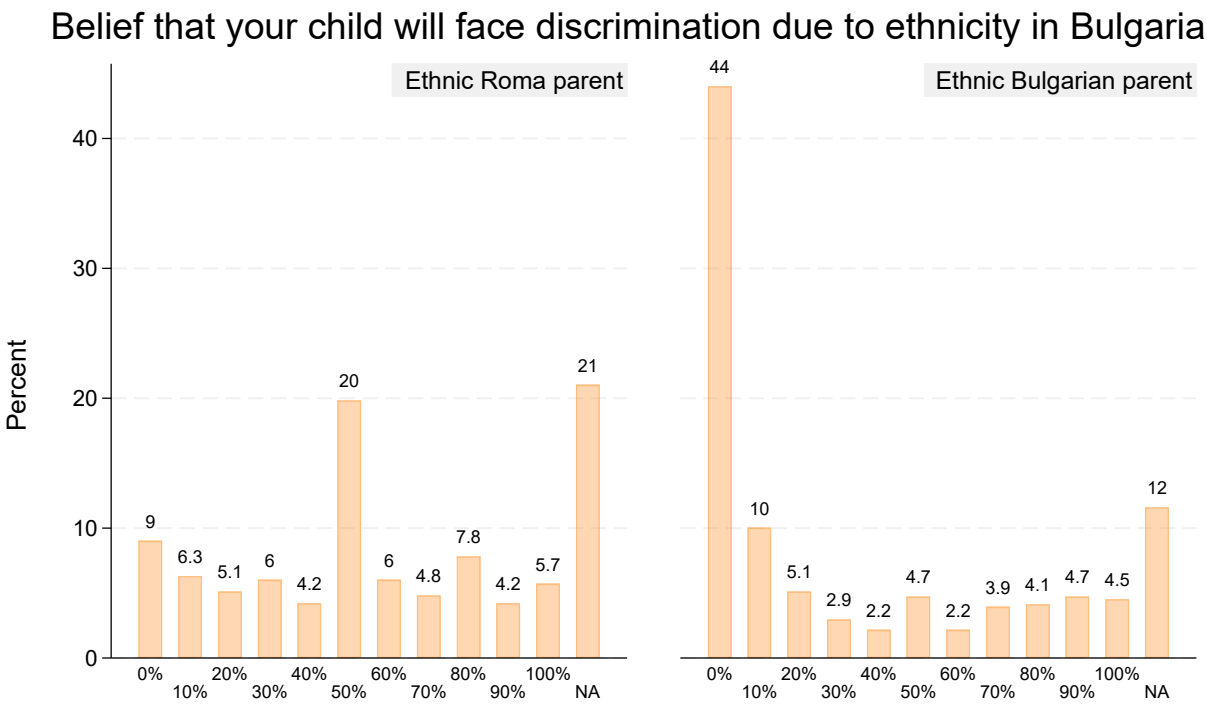


Figure D.3: Expectation of discrimination

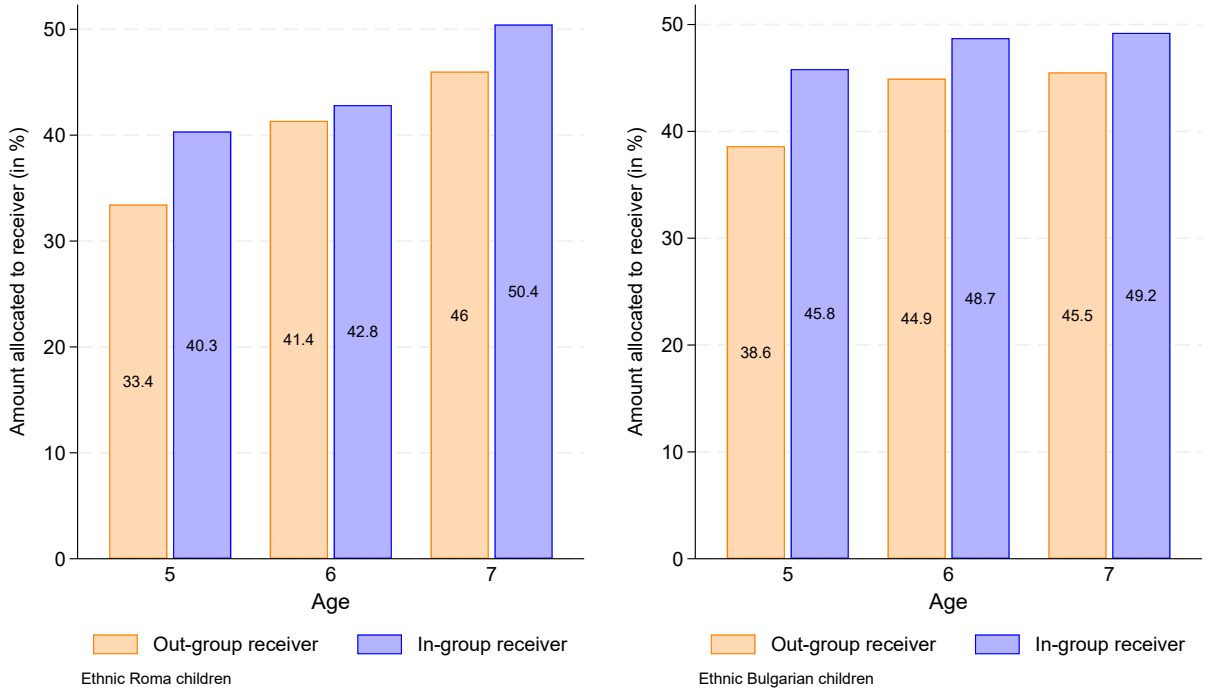


Figure D.4: Shared amount with receivers by children's age

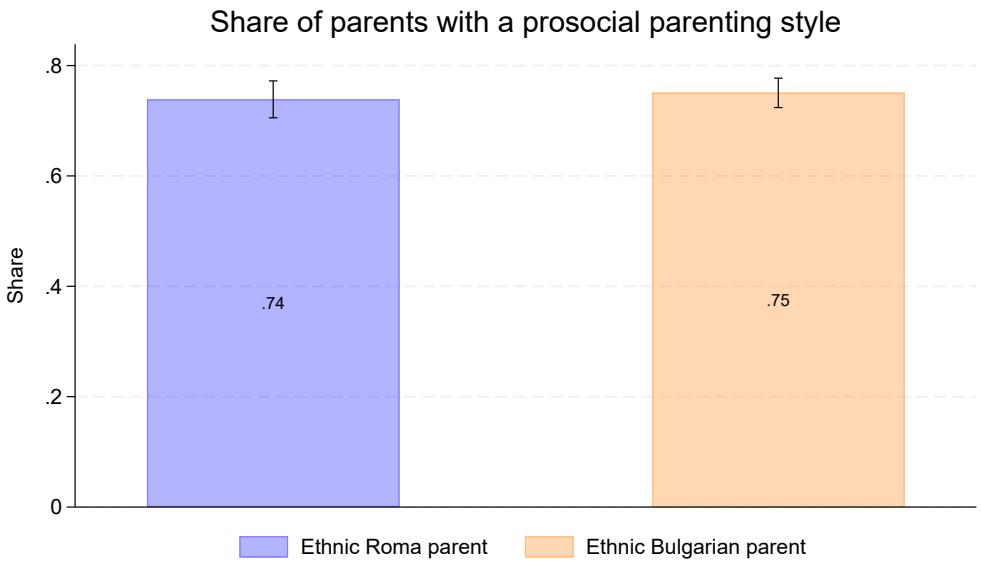


Figure D.5: Prosocial parenting styles

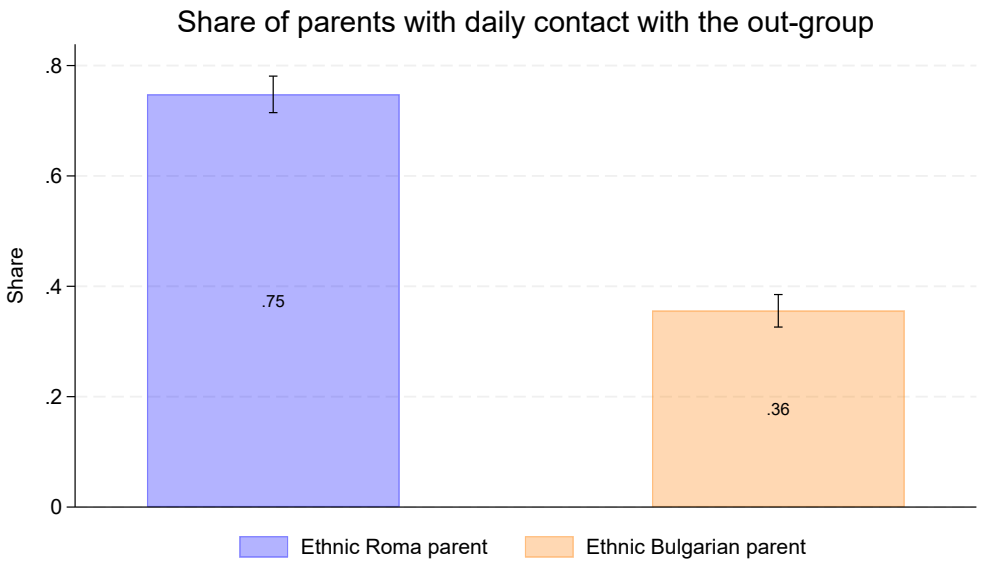


Figure D.6: Frequency of contact with the out-group

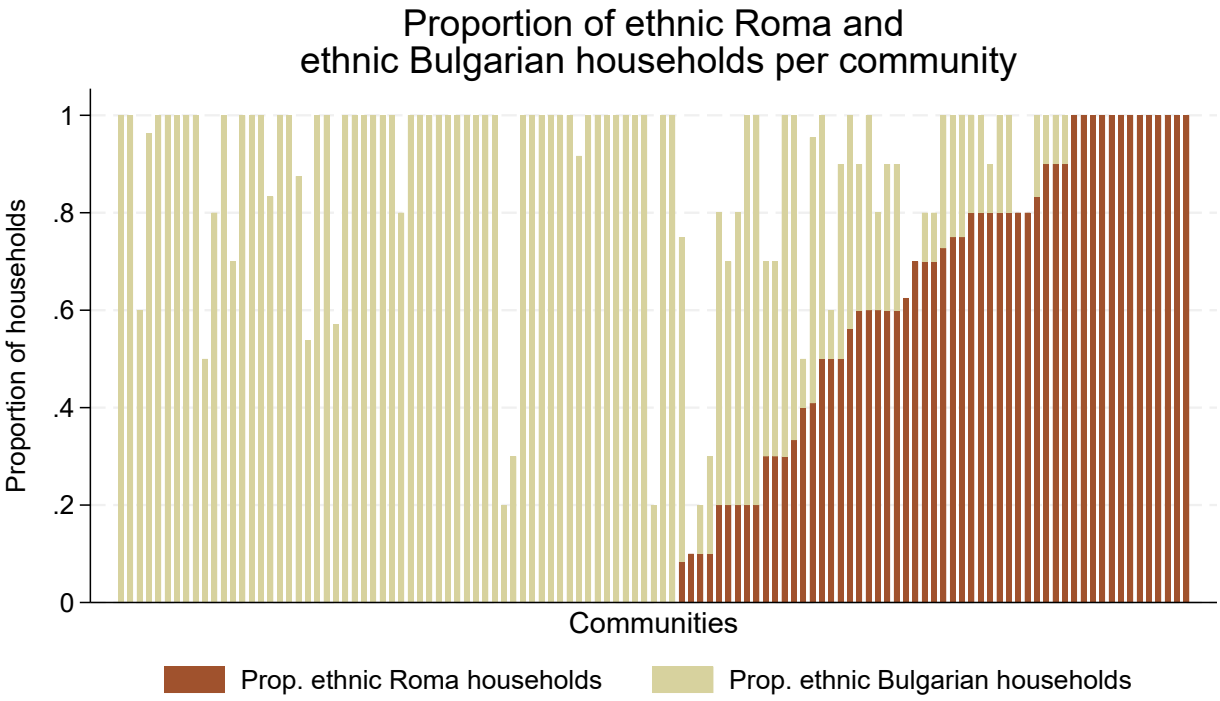


Figure D.7: Proportion of ethnic Roma and ethnic Bulgarian households per community

Table D.1: Survey-based results

	(1) Trust in others - general	(2) Trust in others - personally	(3) Trust in others - first time	(4) Trust in others - religion	(5) Trust in others - ethnicity	(6) Donation to charity (num.)	(7) Willingness to donate
Ethnic Bulgarian parent	0.129**	0.570	-0.200	-0.444	-0.595**	91.468***	1.410***
	(0.060)	(0.396)	(0.252)	(0.295)	(0.289)	(27.579)	(0.296)
Constant	0.135*	7.031***	3.647***	3.951***	4.290***	211.797***	5.292***
	(0.069)	(0.362)	(0.339)	(0.440)	(0.471)	(32.219)	(0.399)
Parents' demographic variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	826	826	826	826	826	826	826

Column 1 is a bivariate variable that equals 1 if the respondent answers that most people can be trusted and 0 if the respondent indicates that most people cannot be trusted; Columns 2, 3 and 4 indicate the level of trust, on a scale from 0 to 10, that the respondent has towards others under certain conditions or demographics (e.g people they know personally, people they met for the first time, people of different religion and people of different ethnicity). Donation to charity indicates the amount each respondent indicated they are willing to donate from a 1,000 leva lottery prize and Willingness to donate indicates their general willingness to donate to good causes without expecting anything in return on a scale from 0 to 10. Majority parent is a bivariate variable that equals 1 if the parent identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma, Turkish or Turkish-Roma. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.2: Intergenerational transmission of social preferences - LPM

	Pooled child's prosociality (median)	Pooled child's prosociality (median)
Pooled parental prosociality (median)	0.154***	0.153***
	(0.046)	(0.047)
Constant	0.509***	0.346
	(0.028)	(0.223)
Children's demographic variables	No	Yes
Parents' demographic variables	No	Yes
Observations	826	826

The table presents the estimations of a linear regression of the amount shared in the dictator game by each child on a variable indicating the degree of pro-sociality of their parent. We pool the amount shared by each child and its parents across their respective rounds of the dictator game and then standardize with respect to their samples. The dependent variable is a bivariate variable that equals one if the amount shared by the child is above the children's sample median and the independent variable is a bivariate variable that equals one if the amount shared by their parent is above the parents' sample median. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.3: Intergenerational transmission of social preferences - above median

	Child's prosociality (pooled)	Child's prosociality (pooled)
Pooled parental prosociality (median)	0.238***	0.234**
	(0.089)	(0.091)
Constant	-0.147***	-0.485
	(0.054)	(0.522)
Children's demographic variables	No	Yes
Parents' demographic variables	No	Yes
Observations	826	826

The table presents the estimations of a linear regression of the amount shared in the dictator game by each child on a variable indicating the degree of pro-sociality of their parent. We pool the amount shared by each child and its parents across their respective rounds of the dictator game and then standardize with respect to their samples. The dependent variable indicates the pooled amount across rounds shared by each child. The independent variable is a bivariate variable that equals one if the amount shared by their parent is above the parents' sample median. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.4: Strength of intergenerational transmission within ethnic groups - interactions

	Child's prosociality
Ethnic Bulgarian child	0.111
	(0.114)
In-group receiver	0.158***
	(0.038)
Parental prosociality	0.080
	(0.079)
In-group receiver × Parental prosociality	0.034
	(0.067)
Ethnic Bulgarian child × Parental prosociality	0.059
	(0.093)
Ethnic Bulgarian child × In-group receiver × Parental prosociality	-0.132
	(0.081)
Constant	0.217
	(0.531)
Children's demographic variables	Yes
Parents' demographic variables	Yes
Observations	1683
Interaction test - receiver	.244
Interaction test - sender	.00287
Interaction test - receiver and sender	.551

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. Ethnic Bulgarian child is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver is from the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. The first interaction test indicates the p-value of the null hypothesis test that the association of a standard deviation increase in parental prosociality, when the receiver is from the in-group, with child's prosociality is zero. The second interaction test indicates the p-value of the null hypothesis test that the association of a standard deviation increase in parental prosociality, when the child is ethnic Bulgarian, with child's prosociality is zero. The third interaction test indicates the p-value of the null hypothesis test that the association of a standard deviation increase in parental prosociality, when the child is ethnic Bulgarian and the receiver is from the in-group, with child's prosociality is zero. Standard errors are clustered at the community level. I include community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.5: Parenting styles and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality
In-group receiver	0.046	0.050
	(0.095)	(0.078)
Parental prosociality	0.201	0.181**
	(0.144)	(0.080)
In-group receiver × Parental prosociality	-0.089	-0.136**
	(0.066)	(0.067)
Prosocial parenting style	0.127	-0.038
	(0.166)	(0.141)
In-group receiver × Prosocial parenting style	0.087	0.179*
	(0.103)	(0.095)
Prosocial parenting style × Parental prosociality	-0.154	-0.052
	(0.140)	(0.074)
In-group receiver × Prosocial parenting style × Parental prosociality	0.156	0.040
	(0.110)	(0.088)
Constant	-0.244	0.260
	(0.728)	(0.566)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018
Prosocial parenting and in-group receiver - estimate	.214	.141
Prosocial parenting and in-group receiver - p-value	.141	.179
Intergenerational association for prosocial parenting - estimate	.0475	.129
Intergenerational association for prosocial parenting - p-value	.579	.00667
Intergenerational association for prosocial parenting with in-group receiver - estimate	.115	.033
Intergenerational association for prosocial parenting with in-group receiver - p-value	.295	.692
Equality of coefficients	.432	.432

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The first column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarian. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver is from the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. Prosocial parenting style is a bivariate variable that equals 1 for those children with parents that present a prosocial parenting style. The first interaction assesses if a prosocial parenting style is associated with child's prosociality when interacting with in-group receivers. The second interaction assesses the intergenerational association estimate for parents with a prosocial parenting style. The third interaction assesses the intergenerational association estimate for parents with a prosocial parenting style and an in-group receiver. Equality of coefficients presents the p-value of the null hypothesis that the association of prosocial parenting style with child's prosociality is the same between ethnic groups. Standard errors are clustered at the community level. I include community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.6: Parental contact with the outgroup and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality
In-group receiver	-0.036	0.200***
	(0.111)	(0.057)
Parental prosociality	0.037	0.171***
	(0.152)	(0.045)
In-group receiver × Parental prosociality	0.120	-0.065
	(0.115)	(0.059)
Daily contact with the out-group	0.062	0.066
	(0.185)	(0.106)
In-group receiver × Daily contact with the out-group	0.211	-0.053
	(0.132)	(0.093)
Daily contact with the out-group × Parental prosociality	0.098	-0.091
	(0.158)	(0.125)
In-group receiver × Daily contact with the out-group × Parental prosociality	-0.170	-0.094
	(0.130)	(0.135)
Constant	-0.180	0.271
	(0.717)	(0.618)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018
Out-group contact and in-group receiver - estimate	.273	.0137
Out-group contact and in-group receiver - p-value	.108	.895
Intergenerational association for out-group contact - estimate	.135	.0801
Intergenerational association for out-group contact - p-value	.109	.476
Intergenerational association for out-group contact and in-group receiver - estimate	.0862	-.0789
Intergenerational association for out-group contact and in-group receiver - p-value	.454	.548
Equality of coefficients	.983	.983

The table presents the estimations of a linear regression of the amount shared in the dictator game on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The first column uses the sample of households that identify as ethnic Roma and the second column uses the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver corresponds to the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. Daily contact with the out-group is a bivariate variable that equals 1 for those children's parents that have daily contact with the ethnic out-group. The first interaction test assesses the null association of daily parental contact with the out-group with increased child's prosociality with in-group receivers. The second interaction test assesses the null association of daily parental contact with the out-group with a stronger association between parental and child's prosociality. Equality of coefficients presents the p-value of the null hypothesis that the association of daily contact with the out-group with child's giving behavior is the same across ethnic groups. Standard errors are clustered at the community level. I include community fixed effects. I control for age and gender of each parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.7: Parental stereotypes and intergenerational transmission of prosociality

	Child's prosociality	Child's prosociality	Child's prosociality	Child's prosociality
In-group receiver	0.119**	0.188***	0.041	0.137**
	(0.058)	(0.049)	(0.071)	(0.053)
Parental prosociality	0.092	0.139***	0.095	0.079*
	(0.076)	(0.048)	(0.092)	(0.040)
In-group receiver × Parental prosociality	0.014	-0.102**	-0.025	-0.043
	(0.062)	(0.049)	(0.079)	(0.055)
Stereotypes towards ethnic Bulgarian	-0.085	0.079		
	(0.099)	(0.048)		
In-group receiver × Stereotypes towards ethnic Bulgarian	0.105	-0.044		
	(0.063)	(0.045)		
Parental prosociality × Stereotypes towards ethnic Bulgarian	-0.023	-0.006		
	(0.109)	(0.039)		
In-group receiver × Parental prosociality × Stereotypes towards ethnic Bulgarian	-0.031	-0.030		
	(0.095)	(0.051)		
Stereotypes towards ethnic Roma			-0.179	0.176***
			(0.140)	(0.053)
In-group receiver × Stereotypes towards ethnic Roma			0.128*	-0.094*
			(0.074)	(0.053)
Parental prosociality × Stereotypes towards ethnic Roma			0.029	-0.076*
			(0.083)	(0.043)
In-group receiver × Parental prosociality × Stereotypes towards ethnic Roma			0.047	0.088
			(0.085)	(0.060)
Constant	-0.055	0.183	0.107	0.138
	(0.752)	(0.576)	(0.753)	(0.619)
Children's demographic variables	Yes	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes	Yes
Sample	Ethnic Roma children	Ethnic Bulgarian children	Ethnic Roma children	Ethnic Bulgarian children
Observations	665	1018	665	1018
Positive stereotypes and in-group receiver - estimate	.0201	.0355	-.0509	.0813
Positive sterotypes and in-group receiver - p-value	.844	.398	.713	.0836
Intergenerational association for positive sterotypes - estimate	.0691	.133	.123	.00291
Intergenerational association for positive sterotypes - p-value	.562	.0812	.186	.959
Intergenerational association for positive sterotypes and in-group receiver - estimate	.144	.059	.145	.0478
Intergenerational association for positive sterotypes and in-group receiver - p-value	.183	.594	.238	.619
Equality of coefficients	.113	.113	.0113	.0113

The table presents the estimations of a linear regression of the amount shared in the dictator game by each child on a set of variables indicating the ethnicity of the receiver and the degree of pro-sociality of their parent. The first and third columns use the sample of households that identify as ethnic Roma and the second and fourth columns use the sample of households that identify as ethnic Bulgarians. The dependent variable indicates the amount, standardized with respect to the whole sample, shared by each child with the receiver during each round of the game. In-group receiver is a bivariate variable that equals 1 for the rounds of the game in which the receiver corresponds to the same ethnic group as the sender. Parental prosociality indicates the amount, standardized with respect to the whole sample, shared by each parent from the same household with the receiver during each round of the game. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. The first interaction assesses if parents with more positive stereotypes are associated with child's prosociality when interacting with in-group receivers. The second interaction assesses the intergenerational association estimate for parents with more positive stereotypes. The third interaction assesses the intergenerational association estimate towards in-group receivers for parents with more positive stereotypes. Equality of coefficients presents the p-value of the null hypothesis that the association of parental stereotypes with child's giving behavior is the same between ethnic groups. We control for age and gender of each child and their parent. Standard errors are clustered at the community level. We include community fixed effects. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.8: Parenting styles and intergenerational transmission of in-group preferences by ethnic group

	Children's in-group preference
Parental in-group preference	0.103
	(0.079)
Ethnic Bulgarian household	-0.157
	(0.404)
Ethnic Bulgarian household × Parental in-group preference	-0.054
	(0.085)
Prosocial parenting style	0.179
	(0.257)
Prosocial parenting style × Parental in-group preference	-0.057
	(0.075)
Ethnic Bulgarian household × Prosocial parenting style	0.179
	(0.329)
Ethnic Bulgarian household × Prosocial parenting style × Parental in-group preference	0.036
	(0.085)
Constant	0.607
	(1.080)
Children's demographic variables	Yes
Parents' demographic variables	Yes
Observations	825
Intergen. assoc. of in-group preference among Roma prosoc. parents - estimate	.0458
Intergen. assoc. of in-group preference among Roma prosoc. parents - p-value	.217
Intergen. assoc. of in-group preference among ethnic Bulgarian prosoc. parents - estimate	.0279
Intergen. assoc. of in-group preference among ethnic Bulgarian prosoc. parents - p-value	.382

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. Prosocial parenting style is an indicator variable that equals 1 for parents that indicate a prosocial parenting style and 0 for parents that do not. The variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among prosocial parents across the whole sample and the different ethnic groups. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.9: Parental contact with the outgroup and intergenerational transmission of in-group preference by ethnic group

	Children's in-group preference
Parental in-group preference	0.071
	(0.051)
Ethnic Bulgarian household	0.612
	(0.472)
Ethnic Bulgarian household × Parental in-group preference	-0.009
	(0.056)
Daily contact with the out-group	0.531
	(0.343)
Daily contact with the out-group × Parental in-group preference	-0.009
	(0.055)
Ethnic Bulgarian household × Daily contact with the out-group	-0.800*
	(0.431)
Ethnic Bulgarian household × Daily contact with the out-group × Parental in-group preference	-0.082
	(0.073)
Constant	0.301
	(1.063)
Children's demographic variables	Yes
Parents' demographic variables	Yes
Observations	825
Intergen. assoc. of in-group preference among Roma parents with daily outgroup contact - estimate	.0618
Intergen. assoc. of in-group preference among Roma parents with daily outgroup contact - p-value	.143
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with daily outgroup contact - estimate	-.0298
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with daily outgroup contact - p-value	.406

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. Daily contact with the out-group is an indicator variable that equals 1 for parents that have daily contact with the out-group and 0 for parents not in daily contact with the out-group. The variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among parents with daily contact with their ethnic out-group across the whole sample and the different ethnic groups. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.10: Parental stereotypes and intergenerational transmission of in-group preferences by ethnic group

	Children's in-group preference	Children's in-group preference
Parental in-group preference	0.078*	0.060
	(0.040)	(0.039)
Ethnic Bulgarian household	0.146	0.028
	(0.393)	(0.319)
Ethnic Bulgarian household × Parental in-group preference	-0.062	-0.026
	(0.048)	(0.048)
Ethnic Roma stereotypes	0.008	
	(0.252)	
Parental in-group preference × Ethnic Roma stereotypes	-0.049	
	(0.048)	
Ethnic Bulgarian household × Ethnic Roma stereotypes	-0.468	
	(0.287)	
Ethnic Bulgarian household × Parental in-group preference × Ethnic Roma stereotypes	0.039	
	(0.052)	
Ethnic Bulgarian stereotypes		0.104
		(0.250)
Parental in-group preference × Ethnic Bulgarian stereotypes		-0.028
		(0.044)
Ethnic Bulgarian household × Ethnic Bulgarian stereotypes		-0.188
		(0.267)
Ethnic Bulgarian household × Parental in-group preference × Ethnic Bulgarian stereotypes		0.023
		(0.047)
Constant	0.531	0.668
	(1.106)	(1.100)
Children's demographic variables	Yes	Yes
Parents' demographic variables	Yes	Yes
Observations	825	825
Intergen. assoc. of in-group preference among Roma parents with more positive stereotypes - estimate	.0298	.0318
Intergen. assoc. of in-group preference among Roma parents with more positive stereotypes - p-value	.576	.589
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with more positive stereotypes - estimate	.00737	.0297
Intergen. assoc. of in-group preference among ethnic Bulgarian parents with more positive stereotypes - p-value	.789	.362

The table presents the linear regression estimates of children's in-group preference on their parents' in-group preference. Ethnic Bulgarian household is a bivariate variable that equals 1 if the child has a parent that identifies as ethnic Bulgarian and equals 0 if the parent identifies as ethnic Roma. The stereotypes variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. The in-group preference variables are not standardized: children were allocated 10 tokens and parents were allocated 20 leva. The interactions test the null hypothesis of no intergenerational association of parental in-group preference among parents with more positive stereotypes towards ethnic Roma children across the different ethnic groups. Standard errors are clustered at the community level. I add community fixed effects. I control for age and gender of each child and their parent. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.11: Expectation of discrimination and child's prosociality

	Child's prosociality (pooled)	Child's prosociality (pooled)	Child's prosociality (pooled)
Expectation above median	0.031	0.076	0.082
	(0.088)	(0.177)	(0.176)
Ethnic Bulgarian child		0.229	0.199
		(0.209)	(0.207)
Expectation above median × Ethnic Bulgarian child		-0.027	-0.039
		(0.196)	(0.195)
Parental prosociality (pooled)			0.099
			(0.060)
Constant	0.952	0.929	0.795
	(0.692)	(0.697)	(0.672)
Children's demographic variables	Yes	Yes	Yes
Parents' demographic variables	Yes	Yes	Yes
Observations	713	713	713
Expected discrimination and ethnic Bulgarian households - estimate		.0484	.0428
Expected discrimination and ethnic Bulgarian households - p-value		.622	.659

Table D.12: Parental stereotypes towards Roma children and daily contact with the out-group

	Stereotypes towards ethnic Roma children	Stereotypes towards ethnic Roma children	Stereotypes towards ethnic Roma children
Daily contact with the out-group	0.581***	0.352***	0.349***
	(0.146)	(0.120)	(0.118)
Ethnic Bulgarian parent	-0.603***	-0.106	-0.106
	(0.157)	(0.200)	(0.197)
Daily contact with the out-group × Ethnic Bulgarian parent	0.044	-0.143	-0.141
	(0.203)	(0.188)	(0.183)
Constant	0.058	-0.084	0.050
	(0.138)	(0.128)	(0.177)
Parents' demographic variables	No	No	Yes
Observations	842	826	826
Community Fixed Effects	N	Y	Y
Positive stereotypes and contact with out-group among ethnic Bulgarian parents - estimate	.626	.209	.207
Positive stereotypes and contact with out-group among ethnic Bulgarian parents - p-value	.0000303	.128	.126

The table presents the estimates of a linear regression model of parental stereotypes on a measure of their daily contact with their ethnic out-group. The dependent variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. Daily contact with the out-group is an indicator variable that equals one for those parents in daily contact with their out-group. Ethnic Bulgarian parent is an indicator variable that equals one for those parents that identify as ethnic Bulgarians. I control for age and gender of each parent. Standard errors are clustered at the community level. Stars indicate statistical significance: * 10% ** 5% *** 1%

Table D.13: Parental stereotypes towards ethnic Bulgarian children and daily contact with the out-group

	Stereotypes towards ethnic Bulgarian children	Stereotypes towards ethnic Bulgarian children	Stereotypes towards ethnic Bulgarian children
Daily contact with the out-group	0.401**	0.228*	0.213
	(0.202)	(0.134)	(0.132)
Ethnic Bulgarian parent	0.254	0.421**	0.402**
	(0.192)	(0.169)	(0.168)
Daily contact with the out-group × Ethnic Bulgarian parent	-0.125	-0.268	-0.249
	(0.233)	(0.185)	(0.180)
Constant	-0.331*	-0.314**	-0.256
	(0.176)	(0.120)	(0.208)
Parents' demographic variables	No	No	Yes
Observations	842	826	826
Community Fixed Effects	N	Y	Y
Positive stereotypes and contact with out-group among ethnic Bulgarian parents - estimate	.276	-.0405	-.0366
Positive stereotypes and contact with out-group among ethnic Bulgarian parents - p-value	.0231	.76	.777

The table presents the estimates of a linear regression model of parental stereotypes on a measure of their daily contact with their ethnic out-group. The dependent variable is a standardized mean index for stereotypes towards children of different ethnic groups in the country. Higher values represent more positive stereotypes. Daily contact with the out-group is an indicator variable that equals one for those parents in daily contact with their out-group. Ethnic Bulgarian parent is an indicator variable that equals one for those parents that identify as ethnic Bulgarians. I control for age and gender of each parent. Standard errors are clustered at the community level. Stars indicate statistical significance: * 10% ** 5% *** 1%

Appendix E Description and descriptive findings of the second dictator game

In the second dictator game, each player is presented with four pre-defined pairs of resource allocations[34]. Each pair contains different allocations of resources between the sender and the receiver, and the player is asked to choose their preferred allocation within each pair[35]. Then, each player's decisions across the four allocations allow us to categorize their type of social preferences. Table E.1 displays the different pre-defined allocations that were presented to parents and table E.2 presents the resulting social preference type depending on the choices made during the game. For example, to clarify how to interpret both tables: an individual would be classified as having altruistic social preferences if, in the costly and costless prosocial games, the individual chooses allocation A, but in the costly and costless envy games chooses allocation B (Bauer et al., 2014).

Table E.1: Allocation choices in the social preference games

	Allocation A		**Allocation B**		**Please circle which allocation you prefer**	
No.	**You receive**	**The other parent receives**	**You receive**	**The other parent receives**		
Costly prosocial game	5 leva	5 leva	10 leva	0 leva	A	B
Costless prosocial game	5 leva	5 leva	5 leva	0 leva	A	B
Costly envy game	5 leva	5 leva	10 leva	15 leva	A	B
Costless envy game	5 leva	5 leva	5 leva	10 leva	A	B

Table E.2: Categorization of social preferences

Types of social preferences				
	Costly prosocial game	**Costless prosocial game**	**Costly envy game**	**Costless envy game**
Altruistic	A	A	B	B
Egalitarian	A	A	A	A
Spiteful	B	B	A	A
Selfish	B	A OR B	B	A OR B

In figures E.2 and E.1, I present the classification of social preferences based on the modified dictator game. First, the data show that more than 80% of the parents can be described in one of the four relevant categories[36] (i.e., altruistic, egalitarian, spiteful, selfish) regardless of receiver's ethnicity. For ethnic Bulgarian parents, around 68% of them display egalitarian social preferences across receivers, while this percentage is around 60% for Roma parents.

[34]This version of the dictator game was originally developed by Fehr et al. (2008) and expanded by Bauer et al. (2014).
[35]The order of the allocations is randomized between subjects.
[36]This percentage is similar to previous results in the literature (Bauer et al., 2014; Chowdhury et al., 2022; Fehr et al., 2008).

Regarding the classification of children's social preferences in figure E.1, the results indicate that there is a large share of children who did not understand the task. Approximately 23% of the ethnic Roma children and around 13% of the ethnic majority children were not able to complete at least one of the rounds in the modified dictator game. This lack of understanding means that we can categorize the social preferences of around 60% of the Roma children and 75% of the ethnic Bulgarian children. Nevertheless, most of the children in both ethnic groups that understood the task have egalitarian social preferences - 45% amongst Roma children and 60% amongst ethnic Bulgarian children.

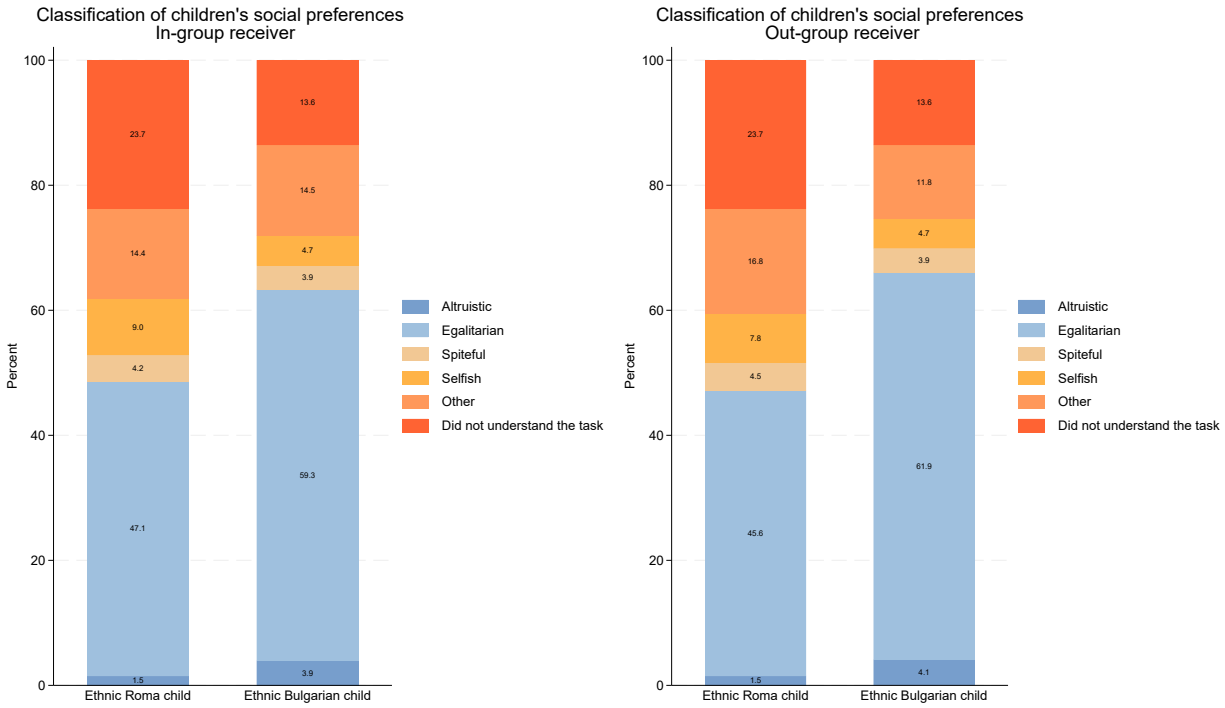


Figure E.1: Classification of children's social preferences

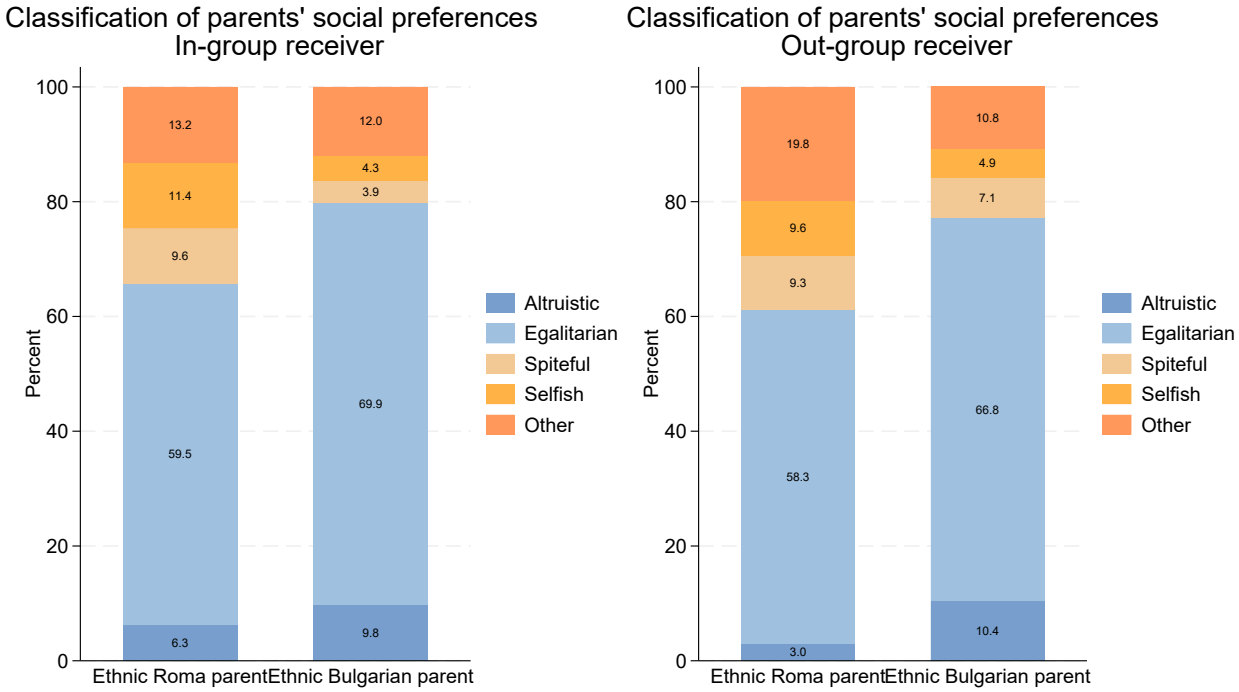


Figure E.2: Classification of parental social preferences